

Methods for exact numerics

A collection of controlled approximations

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Abstract

1 Introduction

Numerical calculations and approximative methods are often viewed as non-exact techniques, used only to build intuition or in situations where exact solutions are not available. This project¹ aims to challenge this simplistic view. Although approximative methods are not designed to provide exact solutions, they can often be supplemented by rigorous bounds. By the careful application of suitable inequalities, together with some global knowledge about the problem at hand, even a finite computation can provide intervals in which the exact solution is guaranteed to lie.

Therefore, by “exact numerics” we mean methods that can provide *exact true statements* about a system after a finite computation. Such statements will often take the form of intervals in which the exact solution, or some quantity of interest, is guaranteed to lie. These exact statements can then be used even in rigorous mathematical proofs. As an example, see the solution of Smale’s 14th problem concerning the Lorenz attractor [74]².

Controlled approximations proved to be fruitful not only in mathematics, but in theoretical physics as well. In condensed matter physics, for example, the proof of the “stability of matter”, given by Lieb and Thirring [44, 45] (first proved by rougher means by Dyson and Lenard [20, 41]), relied heavily on controlled estimates and the careful use of rigorous inequalities³.

The methods collected below do not work, and could not work, for arbitrary problems. Some kind of global knowledge about the problem is necessary in order to obtain meaningful finite enclosing intervals. However, we believe that a sufficiently rich family of numerical problems can be identified and matched with suitable methods that provide such exact statements and are relevant for science, mathematics, and engineering.

Further, it seems possible to develop custom-made exact numerical methods for specific problems. Such problems may not be covered by broad families, or may require finer bounds for a more effective development cycle.

In general, in this work we aim to keep computational methods both rigorous and relevant to their intended applications. We try to provide answers that are not only mathematically controlled, but also useful in practice and suitable for further calculations or decision making. A crucial question that a user of a specific numerical method might ask is: “How close are the results of the approximative method to the exact solution?” In this paper we collect methods by which this naturally arising question can be answered through statements of the form: “The result is closer to the exact solution than an exactly calculable value.”

However this work does not wish to undermine uncontrolled numerical methods. Such methods are often very powerful, and many techniques that are now rigorously understood and controlled started as experimental or uncontrolled approaches. While acknowledging the importance of such methods, including techniques developed and used, for instance, by Bender⁴ and others [9], we want to draw attention to a complementary branch of approximative computation, exact or rigorous numerics.

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¹The manuscript, examples, and related project material are available at github.com/Konczer/ExactNumerics

²For more examples see for instance the talk of Ralph Baker Kearfott or works awarded by the Moore Prize.

³Including Sobolev inequalities [72, 43, 42] and spectral inequalities [40].

⁴See also Carl Bender’s lecture series *Mathematical Physics*.

1.1 Dialogues On Mathematics

As a mathematical and cultural reference, consider the following fictional dialogue between Archimedes and Hieron, king of Syracuse:

Hieron But are strict proofs in applied mathematics really necessary? After all, as you said, the mathematical model is only an approximation of reality. If you use approximately correct formulae, your results will be still approximately close to reality and they can never be absolutely correct anyway.

Archimedes You are mistaken, my king. Just because the mathematical model is only an approximation and there is always a certain discrepancy with the facts, one has to take care not to increase this discrepancy further by a careless use of mathematics. One has to be as accurate as possible. By the way, in regard to approximations, there is a common misunderstanding that using approximations means departing from mathematical precision. Approximations have a precise theory, and results about approximations—for instance, inequalities—have to be proved as rigorously as identities. Perhaps you remember the approximations which I gave for the area of a circle with given diameter; I proved them with a rigor usual in geometry.

—Alfréd Rényi, Dialogues On Mathematics [63, 64]

1.2 A priori and a posteriori certification

Numerical certification can be roughly divided into two flavours: “a priori certification” and “a posteriori certification”. A typical tool for a priori error control is *interval arithmetic*. It has an abundant literature [53, 56, 75], as well as available libraries in several popular programming languages. Examples include the *Interval Arithmetic Library* in C++, *PyInterval* in Python, *interval* in GNU Octave, and *Interval Arithmetic* in Wolfram Mathematica. Interval arithmetic is also covered by an *IEEE standard*.

This work on *numerical certification* focuses mainly on methods of “a posteriori” certification. Their main feature is that the guarantees are not derived from the details of the approximation method itself. Instead, they are determined from a proposed approximation together with some global properties of the problem in question.

The scope considered here goes beyond the control of rounding errors and the study of condition numbers. We are interested more broadly in exact bounds and certified statements obtained from structural and global properties of the underlying problem.

The reliance on inequalities leaves considerable room for creativity without losing rigor. Different verification methods may be better or worse suited to different approximative computations. In particular, the certified enclosure obtained by one method may be tighter or looser than that obtained by another. Nevertheless, as long as the assumptions of the verification method are satisfied, each resulting enclosure provides an exact and provably true statement.

1.3 Broader vision

Besides improving the image and reputation of numerical calculations as methods capable of producing rigorous mathematical statements, exact numerics may also have a broader role in modern computational pipelines, giving a current relevance to the project.

In particular, exact validation methods – especially *a posteriori* methods – can provide an independent test of results obtained by experimental or otherwise uncontrolled computational methods. The method producing a candidate solution could be numerical, heuristic, perturbative, based on machine learning, or even treated as a black box. If the resulting candidate can subsequently be certified against a given specification, then the final result does not have to rely on the internal reliability or “interpretability” of the method that produced it.

A related idea has become increasingly important in pure mathematics. Large language models (LLMs) and other automated methods can be used to propose arguments or formal proofs, while proof assistants such as *Lean* or *Coq* can independently verify the resulting formal statements. This combination of experimental or generative methods with a rigorous verification layer has been advocated and used, among others, by Terence Tao in machine-assisted mathematics [71]. The analogous idea considered here is to provide such a verification layer for numerical computation.

The envisioned pipeline can be divided into three main stages: *specification*, *computation*, and *validation*. First, the numerical problem is specified together with the quantities of interest, the available global properties, and the required tolerances. A computational method then produces a candidate solution. Finally, an independent certification method determines whether the candidate satisfies the required

specification. If it does, the output is a certified result. If it does not, the candidate can be refined or another computational or validation method can be applied.

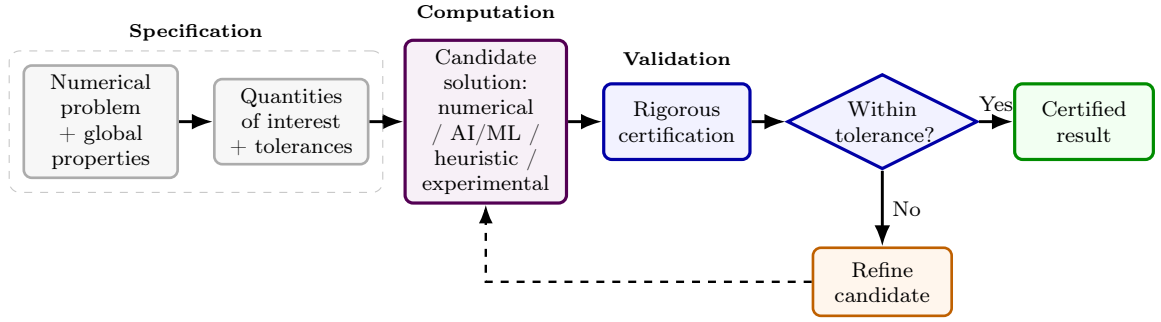


Figure 1: A proposed computational pipeline for exact numerics. The candidate solution may be produced by an approximate, heuristic, or experimental method. An independent certification stage determines whether the quantities of interest satisfy the required tolerances.

One could ask: if the method used to generate the candidate is largely irrelevant to the final guarantee, why not simply guess randomly until a candidate passes the validation test? The reason for retaining sophisticated, and potentially expensive, approximative solvers is that verification may itself be difficult to achieve. A good approximative method can produce candidates that are much more likely to admit tight and successful certification. Moreover, even when verification requires computational effort comparable to that of generating the approximate solution, in many applications this additional cost may be justified by the certainty provided by the final certification.

Such a synergy between computation and validation is particularly useful when the computational method is difficult to analyse directly. A powerful but uncontrolled method can be used to search for a good candidate, while the final guarantee is supplied independently. In this way, experimental methods can be made reliable and AI related results can be cured from hallucinations. Systematic errors only slow down the process, but can't spoil the final certified result. Therefore the output becomes suitable for use in subsequent calculations, mathematical arguments, or decision making.

1.3.1 Certificates and validation protocols

To certify a numerical result, the validator usually needs more information than only the value of the quantity of interest. The computational method can therefore be asked to return, together with its proposed answer, additional data that serves as a *certificate* or *witness* [49] for the validation procedure.

For example, an approximate eigenvalue by itself may not contain enough information for a useful certification. A Krylov–Weinstein bound can instead use an approximate eigenvector together with the corresponding approximate eigenvalue to obtain a rigorous eigenvalue enclosure. Similarly, in a matrix diagonalisation problem one may ask the computational method to return the proposed basis vectors, or the corresponding change of basis, and then analyse the transformed matrix using Gershgorin-type bounds.

This suggests viewing exact numerics not only as a collection of inequalities, but also as the design of *validation protocols*: the “guesser” produces a proposed solution together with suitable auxiliary information, while the validator checks whether these data imply the claimed bounds on the quantities of interest.

There is a loose analogy here with certificate verification in computational complexity and with certain ideas from cryptography, e.g. one-way functions [2, 26]. Producing the relevant solution data may be computationally difficult, while checking a sufficiently informative proposed solution can be substantially easier. The aim is therefore not to reproduce the original computation, but to construct a verification procedure that can efficiently determine whether the proposed result is correct to the required certified accuracy.

1.4 A few simple illustrative examples

For some, it might still be a strange concept to give exact upper and lower bounds for a solution without knowing the precise solution. However, it is possible to give such bounds – which are not necessarily the tightest possible, but nevertheless provide exact and true statements – based only on an approximate solution and, typically, some global property of the problem.

1.4.1 A simple root guessing problem

To demonstrate a basic exact-bound reasoning in the context of root finding, consider the following function and root-finding problem⁵:

$$f(x) = x + \tanh(x), \quad f(x^*) = 1. \quad (1)$$

Let x^* denote the exact solution. Let's imagine that we got a hint that the root is close to $x^\bullet = 0.5$. Is there a way to say something exact about the accuracy of the guess without calculating the root? The answer is affirmative, because we can deduce a global property of the function f . It is easy to see that

$$\forall x \in \mathbb{R} \quad f'(x) \in [1, 2] \quad (2)$$

Therefore, using the mean value theorem [65], we can make the following exact statement:

$$f(x^*) - f(x^\bullet) = f'(\xi)(x^* - x^\bullet), \quad (3)$$

where ξ lies between $x^\bullet$ and x^* . After taking the absolute value of both sides, we can state the following inequality:

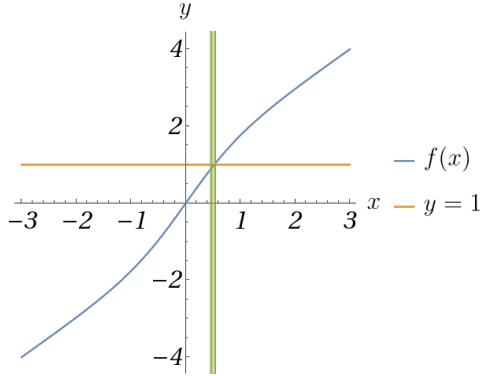
$$|f(x^*) - f(x^\bullet)| = |f'(\xi)| |x^* - x^\bullet| \geq \inf_{\xi \in \mathbb{R}} |f'(\xi)| |x^* - x^\bullet|. \quad (4)$$

Since $\inf_{\xi \in \mathbb{R}} |f'(\xi)| = 1$ for this particular function, we obtain the following exact bound:

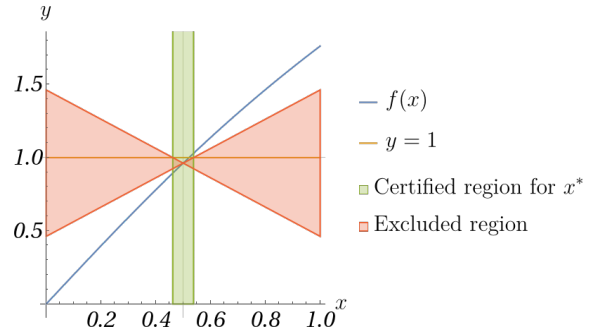
$$|x^* - x^\bullet| \leq \frac{|f(x^*) - f(x^\bullet)|}{\inf_{\xi \in \mathbb{R}} |f'(\xi)|} = |1 - f(x^\bullet)| = |1 - (0.5 + \tanh(0.5))| \leq 0.037883. \quad (5)$$

Therefore, with this educated (or lucky) guess, we can exactly state that

$$x^* \in [0.462117, 0.537883], \quad \text{or equivalently} \quad x^* = 0.5 \pm 0.037883. \quad (6)$$



(a) Certified interval for x^* , $f(x^*) = 1$, based on the approximate guess $x^\bullet = 0.5$.



(b) Closer view, showing the excluded region based on a global lower bound of the derivative of $f(x)$ (specifically $|f'(x)| \geq 1$)

Figure 2: Illustration of a simple “a posteriori” solution-certification example for $f(x) = x + \tanh(x)$. A good approximate guess $x^\bullet = 0.5$, together with a global lower bound on $|f'(x)|$, gives a certified interval containing the exact solution x^* for the equation $f(x^*) = 1$.

1.4.2 Newton's method with exact bounds

Suppose we have a function $f \in C^2(\mathbb{R}, \mathbb{R})$. In fact, it is enough to assume that f is twice differentiable and that the second derivative is bounded on the domain under consideration⁶. We search for a root of this function,

$$x^* \in \mathbb{R}, \quad f(x^*) = 0. \quad (7)$$

Suppose that on some open interval $\mathcal{D}$, containing a point x° , the absolute value of the second derivative can be bounded:

$$\exists M \geq 0, \quad \forall x \in \mathcal{D}, \quad |f''(x)| \leq M. \quad (8)$$

⁵Example (ix) in [56].

⁶More generally, it is enough to assume that $f'(\cdot)$ is locally Lipschitz, with a known Lipschitz constant.

Then, if we are close enough to a root, an exact interval containing a root can be determined. Let us say that we are at x° , where

$$f(x^\circ) \neq 0, \quad f'(x^\circ) \neq 0, \quad (9)$$

and define the corresponding Newton step

$$x^\bullet = x^\circ - \frac{f(x^\circ)}{f'(x^\circ)}. \quad (10)$$

By recalling Taylor's theorem [65], for any x such that the interval between x° and x lies in $\mathcal{D}$,

$$f(x) = f(x^\circ) + f'(x^\circ)(x - x^\circ) + \frac{1}{2}f''(\xi)(x - x^\circ)^2, \quad \xi \in [x^\circ, x]. \quad (11)$$

Therefore,

$$|f(x) - f(x^\circ) - f'(x^\circ)(x - x^\circ)| \leq \frac{1}{2}M(x - x^\circ)^2. \quad (12)$$

By assuming that $f(x) = 0$, i.e. $x = x^*$, we obtain the necessary condition

$$|f(x^\circ) + f'(x^\circ)(x - x^\circ)| \leq \frac{1}{2}M(x - x^\circ)^2. \quad (13)$$

Accordingly, define the candidate set

$$\mathcal{C} = \left\{ x \in \mathbb{R} \mid |f(x^\circ) + f'(x^\circ)(x - x^\circ)| \leq \frac{1}{2}M(x - x^\circ)^2 \right\}. \quad (14)$$

Every root $x^* \in \mathcal{D}$ must belong to $\mathcal{C}$.

Let

$$\mathcal{S} = \text{Comp}_{x^\bullet}(\mathcal{C}) \quad (15)$$

denote the connected component of $\mathcal{C}$ containing the Newton step $x^\bullet$.

If

$$2M|f(x^\circ)| < f'(x^\circ)^2 \quad (16)$$

and

$$\mathcal{S} \subseteq \mathcal{D}, \quad (17)$$

then

$$\exists x^* \in \mathcal{S}, \quad f(x^*) = 0. \quad (18)$$

In the one-dimensional case we can provide explicit lower and upper bounds from this observation simply by solving the second-order inequality

$$|f(x^\circ) + f'(x^\circ)(x - x^\circ)| \leq \frac{1}{2}M(x - x^\circ)^2. \quad (19)$$

For

$$2M|f(x^\circ)| < f'(x^\circ)^2, \quad (20)$$

the connected component containing $x^\bullet$ can be parameterised as

$$I^\circ = \left\{ x^\circ + \frac{f'(x^\circ)}{m} \left(\sqrt{1 - 2m \frac{f(x^\circ)}{f'(x^\circ)^2}} - 1 \right) \mid m \in [-M, 0) \cup (0, M] \right\} \cup \left\{ x^\circ - \frac{f(x^\circ)}{f'(x^\circ)} \right\}. \quad (21)$$

Here the value at $m = 0$ is understood by continuity and is precisely the Newton step $x^\bullet$.

For the particular case

$$f(x^\circ) > 0, \quad f'(x^\circ) > 0, \quad (22)$$

the interval takes the explicit form

$$I^\circ = \left[x^\circ - \frac{f'(x^\circ)}{M} \left(1 - \sqrt{1 - 2M \frac{f(x^\circ)}{f'(x^\circ)^2}} \right), x^\circ - \frac{f'(x^\circ)}{M} \left(\sqrt{1 + 2M \frac{f(x^\circ)}{f'(x^\circ)^2}} - 1 \right) \right]. \quad (23)$$

The Taylor bound gives opposite signs, or a zero, at the two endpoints of this interval. Hence, provided that $I^\circ \subseteq \mathcal{D}$, continuity guarantees the existence of a root

$$x^* \in I^\circ. \quad (24)$$

For $M = 0$, the function is affine on $\mathcal{D}$ and the Newton step $x^\bullet$ is itself the exact root.

This reasoning gives the scalar a posteriori estimate underlying the Newton–Kantorovich theorem [59].

A note on determining the upper bound: In general, determining a valid upper bound of the second derivative of a function on a domain $\mathcal{D}$ can be challenging. However, there are a few simple techniques by which this can be done if the function $f(\cdot)$ is constructed from simpler components.

Let us define a symbolic algebra

$$\mathcal{A} = \langle \mathbb{R}, \cdot, \oplus, \otimes, \mathbf{a}_1, \mathbf{a}_2, \dots \rangle, \quad (25)$$

where an atomic symbol $\mathbf{a}_i$ corresponds to an “elementary function” ϕ_i :

$$\text{Eval}[\mathbf{a}_i](x) = \phi_i(x). \quad (26)$$

For composite algebra elements $A \in \mathcal{A}$, define

$$\text{Eval}[c \cdot A](x) = c \text{Eval}[A](x), \quad (27)$$

$$\text{Eval}[A_1 \oplus A_2](x) = \text{Eval}[A_1](x) + \text{Eval}[A_2](x), \quad (28)$$

and

$$\text{Eval}[A_1 \otimes A_2](x) = \text{Eval}[A_1](x) \text{Eval}[A_2](x). \quad (29)$$

For every $n \in \mathbb{N}$, introduce an upper-bound mapping

$$\overline{B}_n : \mathcal{A} \mapsto \mathbb{R}_{\geq 0}, \quad (30)$$

such that, for the atomic symbols,

$$\overline{B}_n[\mathbf{a}_i] \geq \sup_{x \in \mathcal{D}} |\phi_i^{(n)}(x)|. \quad (31)$$

The bounds can be propagated through scalar multiplication by

$$\overline{B}_n[c \cdot A] = |c| \overline{B}_n[A], \quad (32)$$

and through addition by

$$\overline{B}_n[A_1 \oplus A_2] = \overline{B}_n[A_1] + \overline{B}_n[A_2]. \quad (33)$$

For multiplication, the Leibniz rule gives

$$\overline{B}_n[A_1 \otimes A_2] = \sum_{k=0}^n \binom{n}{k} \overline{B}_k[A_1] \overline{B}_{n-k}[A_2]. \quad (34)$$

For example,

$$\overline{B}_0[A_1 \otimes A_2] = \overline{B}_0[A_1] \overline{B}_0[A_2], \quad (35)$$

$$\overline{B}_1[A_1 \otimes A_2] = \overline{B}_1[A_1] \overline{B}_0[A_2] + \overline{B}_0[A_1] \overline{B}_1[A_2], \quad (36)$$

and

$$\begin{aligned} \overline{B}_2[A_1 \otimes A_2] &= \overline{B}_2[A_1] \overline{B}_0[A_2] + 2 \overline{B}_1[A_1] \overline{B}_1[A_2] \\ &\quad + \overline{B}_0[A_1] \overline{B}_2[A_2]. \end{aligned} \quad (37)$$

Therefore, if for every elementary symbol $\mathbf{a}_i$ the bounds

$$\overline{B}_n[\mathbf{a}_i] \geq \sup_{x \in \mathcal{D}} |\phi_i^{(n)}(x)| \quad (38)$$

are known, then for every composite expression $A \in \mathcal{A}$ the recursively determined value satisfies

$$\overline{B}_n[A] \geq \sup_{x \in \mathcal{D}} \left| \frac{d^n}{dx^n} \text{Eval}[A](x) \right|. \quad (39)$$

In particular, $\overline{B}_2[A]$ provides a valid value of M for the Newton bound above.

This construction may be viewed as a scalar version of recursive derivative inclusion arithmetic, in which interval enclosures of derivatives are replaced by upper bounds on their absolute values [56, Sec. 2.2].

Example: Let

$$\mathcal{D} = (-1, 1) \quad (40)$$

and

$$f(x) = \exp(x) - \arctan(x). \quad (41)$$

One possible symbolic representation is

$$A = \exp \oplus ((-1) \cdot \arctan), \quad \text{Eval}[A](x) = f(x). \quad (42)$$

Since

$$\overline{B}_2[\exp] = e \quad (43)$$

and

$$\overline{B}_2[\arctan] = \frac{3\sqrt{3}}{8}, \quad (44)$$

we obtain

$$\overline{B}_2[A] = e + \frac{3\sqrt{3}}{8} \geq \sup_{x \in \mathcal{D}} |f''(x)|. \quad (45)$$

1.4.3 Multivariate case

The concept of exact solution sets is much more interesting in the multivariate case. In one dimension, for continuous problems, it is often enough to find two points with opposite signs to guarantee a set containing a root.

The [Kantorovich theorem](#), or Newton–Kantorovich theorem [58], provides a corresponding framework in several dimensions.

Newton–Fourier method Another way to obtain exact error bounds is by a modification of the [Fourier method](#) [3, 60].

There are multidimensional extensions as well [62], where solutions are enclosed by nested n -dimensional order intervals (axis-aligned hyperrectangles rather than norm balls).

Newton–Gauss–Seidel interval methods See for example [36].

2 Moderately powerful techniques

We will demonstrate the simpler techniques on the quantum pendulum:

$$H = -\frac{1}{2} \frac{\partial^2}{\partial \varphi^2} + g(1 - \cos(\varphi)) \quad (46)$$

2.1 Gershgorin circle

[Gershgorin](#) or [Geršgorin](#) circle theorem [76, 50, 27, 34] was originally stated for finite complex matrices, and has a refined, stronger form:

Let $A = (a_{ij}) \in \mathbb{C}^{n \times n}$, and define the Gershgorin discs

$$\mathcal{G}_i = \left\{ z \in \mathbb{C} \left| |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right. \right\}, \quad i = 1, \dots, n. \quad (47)$$

Then every eigenvalue of A lies in the union

$$\sigma(A) \subseteq \bigcup_{i=1}^n \mathcal{G}_i. \quad (48)$$

Moreover, if the union of some k Gershgorin discs is disjoint from the union of the remaining $n - k$ discs, then this separated union contains exactly k eigenvalues of A , counted with algebraic multiplicity⁷. In particular, if all Gershgorin discs are mutually disjoint, then each disc contains exactly one eigenvalue.

⁷Algebraic multiplicity can differ from geometric multiplicity in general, it can be larger than the number of linearly independent eigenvectors associated with the eigenvalue [34, 27].

In infinite dimensions the statement needs a little more care. The simple statement about inclusion can be generalised for the eigenvalues and eigenvectors of operators on ℓ^p [68]. Under stronger assumptions, in particular uniform strict diagonal dominance,

$$\exists \sigma \in [0, 1) \quad \text{such that} \quad \sum_{j \neq i} |a_{ij}| \leq \sigma |a_{ii}| \quad \text{for all } i, \quad (49)$$

or the analogous condition for the columns, together with conditions ensuring a discrete spectrum, the statement concerning multiplicities can also be recovered. Namely, if the union of k Gershgorin discs is disjoint from all the remaining discs, then this union contains eigenvalues with total algebraic multiplicity k [21].

2.2 Brauer's ovals of Cassini

Brauer's ovals of Cassini can be viewed as a refinement of the Gershgorin disks [76], giving tighter bounds for the actual eigenvalues.

Brualdi's theorem [15, 37, 16], maybe with non-unitary transformations.

3 Quartic oscillator as a testing ground

Several normalisations of the quartic oscillator are used in the literature. We shall distinguish three conventions. The λ -normalisation, which is close to the field-theoretic convention [73, 61] for a $\lambda\phi^4$ interaction [70], is

$$H_\lambda = \frac{1}{2}P^2 + \frac{1}{2}\omega_0^2 X^2 + \lambda \frac{1}{4!}X^4, \quad (50)$$

The g -normalisation, common in quantum-mechanical treatments [28, 51, 39] of the anharmonic oscillator [17, 79], is

$$H_g = \frac{1}{2}P^2 + \frac{1}{2}\omega_0^2 X^2 + gX^4, \quad (51)$$

Finally, a rescaled form, particularly convenient in mathematical treatments [69, 35], is

$$H_\kappa = P^2 + \omega_0^2 X^2 + \kappa X^4. \quad (52)$$

	H_λ	H_g	H_κ
$H_\lambda(\omega_0, \lambda)$	$H_\lambda(\omega_0, \lambda)$	$H_g(\omega_0, g = \lambda/24)$	$\frac{1}{2}H_\kappa(\omega_0, \kappa = \lambda/12)$
$H_g(\omega_0, g)$	$H_\lambda(\omega_0, \lambda = 24g)$	$H_g(\omega_0, g)$	$\frac{1}{2}H_\kappa(\omega_0, \kappa = 2g)$
$H_\kappa(\omega_0, \kappa)$	$2H_\lambda(\omega_0, \lambda = 12\kappa)$	$2H_g(\omega_0, g = \kappa/2)$	$H_\kappa(\omega_0, \kappa)$

Table 1: Conversion between common normalisations of the quartic oscillator. Each row is expressed in the convention indicated by the column.

4 Controlled approximation

4.1 Majorization, minorization, envelope method

Lemma:

$$\forall x \in X, f(x) \geq g(x) \implies \inf_{x \in X} f(x) \geq \inf_{x \in X} g(x) \quad (53)$$

$$\forall x \in X, f(x) \geq g(x) \implies \sup_{x \in X} f(x) \geq \sup_{x \in X} g(x) \quad (54)$$

Proof for inf:

$$\forall x \in X, f(x) \geq g(x) \geq \inf_{x \in X} g(x) \quad (55)$$

Therefore $\inf_{x \in X} g(x)$ is a global lower bound for $f(\cdot)$, so $\inf_{x \in X} f(x) \geq \inf_{x \in X} g(x)$

Lemma:

$$\forall x \in X', f(x) \geq g(x), X' \subseteq X \implies \inf_{x \in X'} f(x) \geq \inf_{x \in X} g(x) \quad (56)$$

Proof for inf:

$$\forall x \in X', f(x) \geq g(x) \geq \inf_{x \in X'} g(x) \geq \inf_{x \in X} g(x) \quad (57)$$

Therefore $\inf_{x \in X} g(x)$ is a global lower bound for $f(\cdot)$, on X' , so $\inf_{x \in X'} f(x) \geq \inf_{x \in X} g(x)$.

For two self-adjoint linear operators A, B on the finite n -dimensional vector complex vectorspace V we have (for Löwner order see [81, 48]):

$$A \succcurlyeq B \implies \lambda_i \geq \mu_i \quad (58)$$

Proof:

$$A \succcurlyeq B \iff \forall v \in V, v^* \cdot A \cdot v \geq v^* \cdot B \cdot v \quad (59)$$

$$\mu_1 = \min_{v \in V, \|v\|=1} v^* \cdot B \cdot v \leq \min_{v \in V, \|v\|=1} v^* \cdot A \cdot v = \lambda_1 \quad (60)$$

$$\lambda_n = \max_{v \in V, \|v\|=1} v^* \cdot A \cdot v \geq \max_{v \in V, \|v\|=1} v^* \cdot B \cdot v = \mu_n \quad (61)$$

$$\lambda_k = \min_{V_k \subseteq V, \text{Dim}(V_k)=k} \max_{v \in V_k, \|v\|=1} v^* \cdot A \cdot v \quad (62)$$

$$\forall V_k \subseteq V \max_{v \in V_k, \|v\|=1} v^* \cdot A \cdot v \geq \max_{v \in V_k, \|v\|=1} v^* \cdot B \cdot v \quad (63)$$

Therefore

$$\lambda_k = \min_{V_k \subseteq V, \text{Dim}(V_k)=k} \max_{v \in V_k, \|v\|=1} v^* \cdot A \cdot v \geq \min_{V_k \subseteq V, \text{Dim}(V_k)=k} \max_{v \in V_k, \|v\|=1} v^* \cdot B \cdot v = \mu_k \quad (64)$$

In the case of linear operators on infinite-dimensional separable Hilbert spaces $\mathcal{H}$, if A and B are densely defined self-adjoint operators and both A and B have a point spectrum (at least for a finite number of low-lying eigenvalues) – technically both operators having a countable set of eigenvalues and eigenvectors below their “essential spectrum” – and their spectrum is bounded from below, then

$$A \succcurlyeq B \implies \lambda_k \geq \mu_k \quad (65)$$

Proof:

$$A \succcurlyeq B \iff \forall \psi \in \mathcal{Q}(A) \cap \mathcal{Q}(B), \langle \psi | A | \psi \rangle \geq \langle \psi | B | \psi \rangle \quad (66)$$

Where $\mathcal{Q}(A) \subseteq \mathcal{H}$ denotes the linear space where the quadratic form is defined, i.e. $\langle \psi | A | \psi \rangle \in \mathbb{R}$.

In fact, in this case – when both A and B are bounded from below and $A \succcurlyeq B$ – $\mathcal{Q}(A) \subseteq \mathcal{Q}(B)$ because for a state $\psi \in \mathcal{Q}(A)$, the quadratic form $\langle \psi | A | \psi \rangle$ is defined; then $\langle \psi | B | \psi \rangle$ has to be finite and defined as well. This implies that $\mathcal{Q}(A) \cap \mathcal{Q}(B) = \mathcal{Q}(A)$ that is a dense subspace of $\mathcal{H}$.

Utiliyng the Min-max theorem (in Theorem 4.10)

$$\lambda_k = \inf_{\mathcal{H}_k \subseteq \mathcal{H}, \text{Dim}(\mathcal{H}_k)=k} \sup_{\psi \in \mathcal{H}_k, \|\psi\|=1} \langle \psi | A | \psi \rangle \quad (67)$$

$$\forall \mathcal{H}_k \subseteq \mathcal{Q}(A) \sup_{\psi \in \mathcal{H}_k, \|\psi\|=1} \langle \psi | A | \psi \rangle \geq \sup_{\psi \in \mathcal{H}_k, \|\psi\|=1} \langle \psi | B | \psi \rangle \quad (68)$$

Therefore:

$$\lambda_k = \inf_{\mathcal{H}_k \subseteq \mathcal{Q}(A), \text{Dim}(\mathcal{H}_k)=k} \sup_{\psi \in \mathcal{H}_k, \|\psi\|=1} \langle \psi | A | \psi \rangle \geq \inf_{\mathcal{H}_k \subseteq \mathcal{Q}(B), \text{Dim}(\mathcal{H}_k)=k} \sup_{\psi \in \mathcal{H}_k, \|\psi\|=1} \langle \psi | B | \psi \rangle = \mu_k \quad (69)$$

In particular, we will show how envelope theory [29] can be formulated as a minorization-majorization technique.

4.1.1 Quartic potential

$$H = \frac{1}{2}P^2 + X^4 \quad (70)$$

$$\check{H}(a) = \frac{1}{2}P^2 + a^4 + 2a^2(X - a)(X + a) \quad (71)$$

$$\check{H}(a) = \frac{1}{2}P^2 + \frac{1}{2}4a^2X^2 - 2a^4 \quad (72)$$

$$\check{E}_n(a) = 2a(n + 1/2) - 2a^4 \quad (73)$$

$$\check{E}_n^* = \sup_{a>0} \check{E}_n(a) \quad (74)$$

$$\check{E}_n^* = \frac{3}{8}(1 + 2n)^{4/3}, \quad a_n^* = \frac{1}{2}(1 + 2n)^{1/3} \quad (75)$$

$$E_n \geq \check{E}_n^* = \frac{3}{8}(1 + 2n)^{4/3} \quad (76)$$

4.1.2 Rectangular potential wells

One of the simplest quantum mechanical systems is a particle in a potential well with infinite walls.

$$V(x; V_0, a) = \begin{cases} V_0 & \text{if } -a < x < a \\ \infty & \text{otherwise} \end{cases} \quad (77)$$

This definition is, of course, far from precise – and this operator is not densely defined on square-integrable functions on the whole number line in $L^2(\mathbb{R}, \mathbb{C})$ –, but it can be defined as a limiting case of densely defined operators.

$$V(x; V_0, a; \Lambda) = \begin{cases} V_0 & \text{if } -a < x < a \\ \Lambda V_{\text{ext}}(x) & \text{otherwise} \end{cases} \quad (78)$$

Where $V_{\text{ext}}(\cdot)$ is a positive continuous function with positive global lower bound $\inf_{x \in \mathbb{R}} V_{\text{ext}}(x) > 0$. If we wish to keep, for instance, the Schwartz functions on the whole number line to belong to the domain of $V(x; V_0, a; \Lambda)$ for any finite $\Lambda > 0$, then we can demand that $V_{\text{ext}}(x)$ has at most polynomial growth as $|x|$ goes to infinity. Quantities related to $V(x; V_0, a)$ can then be understood as the $\Lambda \rightarrow \infty$ limits of expressions involving the regularised potential $V(x; V_0, a; \Lambda)$.

$$\psi_n(x) = \frac{1}{\sqrt{a}} \sin(k_n(x + a)), \quad E_n = \frac{1}{2}k_n^2 + V_0 \quad (79)$$

$$k_n = \pi \frac{n + 1}{2a} \quad (80)$$

4.1.3 Pöschl–Teller potential

See [22].

4.2 Krylov–Weinstein bound

For references see [82, 83, 80, 18, 55], presenting as the special case of Bauer–Fike theorem [66].

For a self-adjoint A

$$\exists v \in V \setminus \underline{0}, \varepsilon \geq 0, v^* \cdot (A - \alpha)^2 \cdot v \leq \varepsilon v^* \cdot v \implies \exists \lambda_i, |\lambda_i - \alpha| \leq \sqrt{\varepsilon} \quad (81)$$

Proof

$$\forall v \in V, v^* \cdot (A - \alpha)^2 \cdot v \geq \min_i (\lambda_i - \alpha)^2 v^* \cdot v \quad (82)$$

$$\exists v \in V \setminus \underline{0}, \varepsilon \geq 0, v^* \cdot (A - \alpha)^2 \cdot v \leq \varepsilon v^* \cdot v \implies \varepsilon \geq \min_i (\lambda_i - \alpha)^2 \quad (83)$$

$$\min_i (\lambda_i - \alpha)^2 \leq \varepsilon \implies \exists \lambda_i, |\lambda_i - \alpha| \leq \sqrt{\varepsilon} \quad (84)$$

Excluding eigenvalues:

$$\exists \varepsilon \geq 0, (A - \alpha)^2 \geq \varepsilon \implies \nexists \lambda_i, \lambda_i \in [\alpha - \sqrt{\varepsilon}, \alpha + \sqrt{\varepsilon}] \quad (85)$$

Proof

Lets assume that for some $\varepsilon > 0$ there is an eigenvalue $\lambda_i \in [\alpha - \sqrt{\varepsilon}, \alpha + \sqrt{\varepsilon}]$. Then there is an eigenvector $v_i \in V \setminus \{0\}$ such that $A \cdot v_i = \lambda_i v_i$. Then $(A - \alpha)^2 \cdot v_i = (\lambda_i - \alpha)^2 v_i$ and

$$\frac{v_i^* \cdot (A - \alpha)^2 \cdot v_i}{v_i^* \cdot v_i} = (\lambda_i - \alpha)^2 \leq \varepsilon \quad (86)$$

Bur from

$$(A - \alpha)^2 \geq \varepsilon \implies \forall v \in V \setminus \{0\}, \frac{v_i^* \cdot (A - \alpha)^2 \cdot v_i}{v_i^* \cdot v_i} \geq \varepsilon \quad (87)$$

This is a contradiction.

$$H = \frac{1}{2}P^2 + V(X) \quad (88)$$

$$(H - E)^2 = \frac{1}{4}P^4 + \frac{1}{2}(P^2 (V(X) - E) + (V(X) - E) P^2) + (V(X) - E)^2 \quad (89)$$

$$[V(X), P] = iV'(X) \quad (90)$$

$$(H - E)^2 = \frac{1}{4}P^4 + P (V(X) - E) P + [[V(X) - E, P], P] + (V(X) - E)^2 \quad (91)$$

$$(H - E)^2 = \frac{1}{4}P^4 + P (V(X) - E) P - V''(X) + (V(X) - E)^2 \quad (92)$$

Minorization:

We can introduce a piecewise constant, minorizing potential $\check{V}(x; E)$, $\forall x \in \mathbb{R}$, $\check{V}(x; E) \leq V(x) - E$ and a minorization square-like potential $\check{W}(X; E)$ $\forall x \in \mathbb{R}$, $\check{W}(x; E) \leq (V(x) - E)^2 - V''(x)$

$$\left(\frac{1}{4}P^2 + P \check{V} P + \check{W} \right) \psi(x) = \varepsilon \psi(x) \quad (93)$$

With the ansatz $\psi(x) = e^{ik \cdot x}$

$$\frac{1}{4}k^4 - \check{V}k^2 + \check{W} = \varepsilon \quad (94)$$

This has four soultions:

$$k_{s_1, s_2} = s_1 \sqrt{2} \sqrt{s_2 \sqrt{\check{V}^2 - \check{W} + \varepsilon} + \check{V}} \quad (95)$$

$$\psi(x) = A_{++} e^{ik_{++}x} + A_{+-} e^{ik_{+-}x} + A_{-+} e^{ik_{-+}x} + A_{--} e^{ik_{--}x} \quad (96)$$

With $A_{++}, A_{+-}, A_{-+}, A_{--} \in \mathbb{C}$

$$(H - E)^2 = \frac{1}{4}P^4 + (V(X) - E) P^2 - [V(X) - E, P]P - V''(X) + (V(X) - E)^2 \quad (97)$$

$$(H - E)^2 = \frac{1}{4}P^4 + (V(X) - E) P^2 - iV'(X)P - V''(X) + (V(X) - E)^2 \quad (98)$$

For piecewise constant $V(x)$, going from V_- for $x < 0$ and V_+ for $x > 0$

$$(H - E)^2 \psi(x) = \frac{1}{4} \psi''''(x) + (V(x) - E)(-\psi''(x)) - i(V_+ - V_-) \delta(x)(-i\psi'(x)) - V''(X) \psi(x) + (V(X) - E)^2 \psi(x) \quad (99)$$

$$(H - E)^2 \psi(x) = \varepsilon \psi(x) \quad (100)$$

$$\int_{-\Delta}^{\Delta} (H - E)^2 \psi(x) dx = \int_{-\Delta}^{\Delta} \varepsilon \psi(x) dx \quad (101)$$

$$\lim_{\Delta \rightarrow 0^+} \int_{-\Delta}^{\Delta} \psi''''(x) dx = \psi_+''' - \psi_-''' \quad (102)$$

$$\lim_{\Delta \rightarrow 0^+} \int_{-\Delta}^{\Delta} (V(x) - E)(-\psi''(x)) dx = \lim_{\Delta \rightarrow 0^+} [(V(x) - E)(-\psi'(x))]_{-\Delta}^{\Delta} - \int_{-\Delta}^{\Delta} V'(x)(-\psi'(x)) dx \quad (103)$$

0 if $\psi'_+ = \psi'_-$

$$\lim_{\Delta \rightarrow 0^+} \int_{-\Delta}^{\Delta} (V_+ - V_-) \delta(x) \psi'(x) dx = (V_+ - V_-) \psi'(0) \quad (104)$$

As $\Delta \rightarrow 0^+$ we get:

$$\frac{1}{4}(\psi_+''' - \psi_-''') + (V_+ - V_-) \psi'(0) = 0 \quad (105)$$

$$\psi_- = \psi_+ = \psi(0), \quad \psi'_- = \psi'_+ = \psi'(0), \quad \psi''_- = \psi''_+ = \psi''(0), \quad (106)$$

4.2.1 Auxiliary space methods

If we have a self-adjoint Hamiltonian operator H on a Hilbert space $\mathcal{H}$, then we can try to find an operator F on an extended Hilbert space $\mathcal{H} \otimes V$ such that F^2 is “close to” $H \otimes I$.

$$H = P^2 + X^4 \quad (107)$$

$$F = P \otimes \sigma_x + W(X) \otimes \sigma_y \quad (108)$$

$$F^2 = P^2 \otimes I + W(X)^2 \otimes I + PW(x) \otimes \sigma_x \sigma_y + W(X)P \otimes \sigma_y \sigma_x \quad (109)$$

$$F^2 = P^2 \otimes I + W(X)^2 \otimes I - i[P, W(x)] \otimes \sigma_z \quad (110)$$

$$F^2 = P^2 \otimes I + W(X)^2 \otimes I - W'(X) \otimes \sigma_z \quad (111)$$

If we chose $W(X) = X^2$, then we get:

$$F^2 = P^2 \otimes I + X^4 \otimes I - 2X \otimes \sigma_z \quad (112)$$

This is not exactly the quartic potential in the original H , but the $-2X \otimes \sigma_z$ term is “small” relative to $P^2 \otimes I + X^4 \otimes I$.

This means that the $x^4 \pm 2x$ potential can be minorized and majorized by $g(x - x_0)^4 + v_0$ potentials; therefore, knowing the eigenvalues for the potential $x^4 \pm 2x$ can provide finite upper and lower bounds for the pure quartic potential.

Using the Krylov-Weinstein bound for F :

Lemma: If we have a self-adjoint A

$$v^* \cdot (A - \sqrt{E})^2 \cdot v \leq \varepsilon v^* \cdot v \quad (113)$$

then there is a λ eigenvalue of A^2 that:

$$\exists \lambda, |\lambda - E| \leq \sqrt{\varepsilon}(2\sqrt{E} + \sqrt{\varepsilon}) \quad (114)$$

Proof

$$v^* \cdot (A - \sqrt{E})^2 \cdot v \leq \varepsilon v^* \cdot v \implies \exists \mu, (\mu - \sqrt{E})^2 \leq \varepsilon \quad (115)$$

$$|\lambda - E| = |\mu - \sqrt{E}| |\mu + \sqrt{E}| \quad (116)$$

$$|\mu + \sqrt{E}| = |\mu - \sqrt{E} + 2\sqrt{E}| \leq |\mu - \sqrt{E}| + 2\sqrt{E} \quad (117)$$

$$|\lambda - E| \leq \sqrt{\varepsilon}(2\sqrt{E} + \sqrt{\varepsilon}) \quad (118)$$

$$(F - \sqrt{E})^2 = F^2 + E - 2\sqrt{E} F \quad (119)$$

$$(F - \sqrt{E})^2 = \begin{bmatrix} P^2 + W(X)^2 - W'(X) + E & 2\sqrt{E}(P + iW(X)) \\ 2\sqrt{E}(P - iW(X)) & P^2 + W(X)^2 + W'(X) + E \end{bmatrix} \quad (120)$$

And whenever we can find a pair of functions $\psi_\uparrow(x), \psi_\downarrow(x)$ that

$$[\psi_\uparrow^*(x), \psi_\downarrow^*(x)] \begin{bmatrix} P^2 + W(X)^2 - W'(X) + E & 2\sqrt{E}(P + iW(X)) \\ 2\sqrt{E}(P - iW(X)) & P^2 + W(X)^2 + W'(X) + E \end{bmatrix} \begin{bmatrix} \psi_\uparrow(x) \\ \psi_\downarrow(x) \end{bmatrix} = \varepsilon \quad (121)$$

then we know that ...

In the quartic case the $x \rightarrow -x$ is a symmetry, therefore we can look for the $(\psi_\uparrow(x), \psi_\downarrow(x))$ pairs as $(\psi(x)/\sqrt{2}, \psi(-x)/\sqrt{2})$

4.2.2 Super symmetric field theory example

The auxiliary space method can be generalised to SUSY field theories, where observable operators – such as H and P – can be defined as the products of some charge operators $\mathcal{Q}$. In the following, we present a sketch of how non-trivial approximative dispersion relations can be derived in 1+1D SUSY field theories.

4.3 Schur-complement method

Splitting the vector space into two component – for example “Low” and “High” energy components.

$$V = V_L \oplus V_H \quad (122)$$

For a self adjoint operator H :

$$H = \begin{bmatrix} H_{LL} & H_{LH} \\ H_{HL} & H_{HH} \end{bmatrix} \quad (123)$$

$$H_{i,j} : V_j \mapsto V_i$$

Lemma: The lowest eigenvalues of H_{LL} μ_i are greater or equal then the lowest eigenvalues of the complete H , λ_i .

Proof With the min-max definition:

$$\lambda_i = \inf_{v_1, \dots, v_i \in V \setminus \{0\}} \max_k \frac{v_k^* \cdot H \cdot v_k}{v_k^* \cdot v_k} \leq \inf_{v_1, \dots, v_i \in V_L \setminus \{0\}} \max_k \frac{v_k^* \cdot H \cdot v_k}{v_k^* \cdot v_k} = \mu_i \quad (124)$$

Because $(V_L \setminus \{0\}) \subseteq (V \setminus \{0\})$

For an eigenvector $v = v_L \oplus v_H$

$$H_{LL} \cdot v_L + H_{LH} \cdot v_H = \lambda v_L \quad (125)$$

$$H_{HL} \cdot v_L + H_{HH} \cdot v_H = \lambda v_H \quad (126)$$

If $(H_{HH} - \lambda)$ is invertible, then

$$v_H = -(H_{HH} - \lambda)^{-1} H_{HL} \cdot v_L \quad (127)$$

And the eigenvalue equation in the lower sector become:

$$H_{\text{eff}}(\lambda) \cdot v_L = \lambda v_L \quad (128)$$

Substituting back to the set of equations in the low energy sector:

$$H_{LL} \cdot v_L - H_{LH}(H_{HH} - \lambda)^{-1} H_{HL} \cdot v_L = \lambda v_L \quad (129)$$

We can introduce an effective matrix $H_{\text{eff}}(\lambda) : V_L \mapsto V_L$

$$H_{\text{eff}}(\lambda) = H_{LL} - H_{LH}(H_{HH} - \lambda)^{-1}H_{HL} \quad (130)$$

If we can determine a lower bound m for H_{HH} that is greater than a low lying eigenvalue, then

$$\|(H_{HH} - \lambda)^{-1}\| \leq (m - \lambda)^{-1} \quad (131)$$

If $\| -H_{LH}(H_{HH} - \lambda)^{-1}H_{HL} \|$ could be made smaller and smaller by expanding V_L , then we could decrease operator norm to obtain upper and lower bounds for the eigenvalues. However even if this operator norm is not decreasing, upper and lower bounds can be obtained for low lying eigenvalues if the eigenvector components are not “scattered too heavily to the high energy sector”.

$$H_{\text{eff}}(B) = H_{LL} - H_{LH} B H_{HL} \quad (132)$$

$$\Lambda = \{(\lambda, v, B) \mid \lambda < m, B \geq 0, \|B\| \leq (m - \lambda)^{-1}, \|v\| = 1, H_{\text{eff}}(B) \cdot v = \lambda v\} \quad (133)$$

Let μ_i, w_i be the set of eigenvalues and eigenvectors of H_{LL} . Then let Λ_i be the connected component of Λ with $(\mu_i, w_i, 0)$. (With possibly $\Lambda_i = \Lambda_j$ for some $i \neq j$.)

We can introduce simple projection operators: $\pi_\lambda(\lambda, v, B) = \lambda$, $\pi_{\lambda v}(\lambda, v, B) = (\lambda, v)$

All $\pi_\lambda(\Lambda_i)$ contain at least one eigenvalue – smaller than m – of the original H matrix acting on $V = V_L \oplus V_H$. Further, all $\pi_{\lambda v}(\Lambda_i)$ contain all the eigenvalues and eigenvectors that belong to the same connected component. $\Lambda_i = \Lambda_j \implies (\lambda_j, v_j) \in \pi_{\lambda v}(\Lambda_i)$

Lemma for eigenvalues $\lambda_i(H_{\text{eff}}(0)) = \lambda_i(H_{LL}) < m$, we have: $\lambda_i(H) \in [\lambda_i(H_{\text{eff}}((m - \lambda_i(H))^{-1}I)), \lambda_i(H_{\text{eff}}(0))]$

4.4 Intermediate-problem methods

See [1, 7, 6, 23, 14]

4.4.1 Bazley’s special choice

See references in [46, 47, 78].

Consider a self-adjoint Hamiltonian

$$H = H_0 + V, \quad (134)$$

where H_0 is exactly solvable and the perturbation is positive,

$$V > 0. \quad (135)$$

Assume, in addition, that V^{-1} exists as a bounded positive operator. Let

$$H_0 \phi_n = \varepsilon_n \phi_n, \quad (136)$$

where the ϕ_n form an orthonormal set of eigenvectors of H_0 .

The general Bazley–Fox construction introduces vectors p_i such that $V p_i$ belongs to a finite-dimensional span of known eigenvectors of H_0 . A particularly simple version of this “special choice” is obtained by taking

$$p_i = V^{-1} \phi_i, \quad (137)$$

so that

$$V p_i = \phi_i. \quad (138)$$

For a finite collection $\phi_1, \dots, \phi_N$, define the Gram matrix

$$G_{ij} = \langle p_i, V p_j \rangle = \langle \phi_i, V^{-1} \phi_j \rangle. \quad (139)$$

Since V^{-1} is positive, G is positive definite provided the selected vectors are linearly independent.

The corresponding finite-rank approximation to V is

$$V_N = \sum_{i,j=1}^N |\phi_i\rangle (G^{-1})_{ij} \langle \phi_j|. \quad (140)$$

The essential property of this construction is

$$0 \leq V_N \leq V. \quad (141)$$

To see this, define

$$q_i = V^{-1/2} \phi_i. \quad (142)$$

Then

$$G_{ij} = \langle q_i, q_j \rangle. \quad (143)$$

Let Π_N denote the orthogonal projector onto

$$\mathcal{Q}_N = \text{span}\{q_1, \dots, q_N\}. \quad (144)$$

Using the standard expression for a projector in a nonorthogonal basis,

$$\Pi_N = \sum_{i,j=1}^N |q_i\rangle (G^{-1})_{ij} \langle q_j|, \quad (145)$$

one finds

$$V_N = V^{1/2} \Pi_N V^{1/2}. \quad (146)$$

Since

$$0 \leq \Pi_N \leq I, \quad (147)$$

it follows immediately, in the quadratic-form sense, that

$$0 \leq V^{1/2} \Pi_N V^{1/2} \leq V^{1/2} I V^{1/2} = V. \quad (148)$$

We may therefore define the intermediate Hamiltonian

$$H_N = H_0 + V_N. \quad (149)$$

Equation (141) gives

$$H_N \leq H. \quad (150)$$

Consequently, the min-max principle implies that the ordered discrete eigenvalues satisfy

$$E_n^{(N)} \leq E_n, \quad (151)$$

where $E_n^{(N)}$ is the n -th eigenvalue of H_N and E_n is the corresponding eigenvalue of H .

The special choice makes the spectrum of H_N particularly easy to determine. Let

$$\mathcal{S}_N = \text{span}\{\phi_1, \dots, \phi_N\}. \quad (152)$$

On this subspace,

$$H_0 \leftrightarrow D_N = \text{diag}(\varepsilon_1, \dots, \varepsilon_N), \quad (153)$$

while

$$V_N \leftrightarrow G^{-1}. \quad (154)$$

Hence

$$\boxed{H_N|_{\mathcal{S}_N} \leftrightarrow D_N + G^{-1}.} \quad (155)$$

Moreover, if ϕ_m is orthogonal to $\mathcal{S}_N$, then

$$V_N \phi_m = 0, \quad (156)$$

and therefore

$$H_N \phi_m = \varepsilon_m \phi_m. \quad (157)$$

Thus the full spectrum of the intermediate Hamiltonian consists of the eigenvalues of the finite matrix $D_N + G^{-1}$, together with the unchanged eigenvalues of H_0 outside the selected subspace. After ordering this combined spectrum, Eq. (151) gives rigorous lower bounds for the corresponding eigenvalues of H .

Example: quartic anharmonic oscillator Consider

$$H = P^2 + X^2 + X^4 + 1, \quad (158)$$

with

$$H_0 = P^2 + X^2, \quad V = X^4 + 1. \quad (159)$$

Since

$$V \geq I, \quad (160)$$

the inverse

$$V^{-1} = \frac{1}{X^4 + 1} \quad (161)$$

is a bounded positive operator.

The normalized eigenfunctions of H_0 are the harmonic-oscillator states

$$\phi_n(x) = \frac{1}{\sqrt{2^n n! \sqrt{\pi}}} H_n(x) e^{-x^2/2}, \quad (162)$$

with eigenvalues

$$\varepsilon_n = 2n + 1. \quad (163)$$

Because the Hamiltonian is parity invariant, the even and odd sectors may be treated independently. In the even sector we choose

$$\phi_0, \phi_2, \dots, \phi_{2(N-1)}. \quad (164)$$

The corresponding special-choice vectors are

$$p_n = \frac{1}{X^4 + 1} \phi_{2n}, \quad n = 0, \dots, N-1, \quad (165)$$

and indeed

$$V p_n = \phi_{2n}. \quad (166)$$

The Gram matrix is therefore

$$G_{mn}^{(+)} = \left\langle \phi_{2m}, \frac{1}{X^4 + 1} \phi_{2n} \right\rangle, \quad (167)$$

or explicitly,

$$G_{mn}^{(+)} = \int_{-\infty}^{\infty} \frac{\phi_{2m}(x) \phi_{2n}(x)}{x^4 + 1} dx. \quad (168)$$

The unperturbed even-sector energies are

$$\varepsilon_{2n} = 4n + 1, \quad (169)$$

so that

$$D_N^{(+)} = \text{diag}(1, 5, 9, \dots, 4N-3). \quad (170)$$

The nontrivial finite block of the Bazley intermediate Hamiltonian is therefore

$$B_N^{(+)} = D_N^{(+)} + \left(G_N^{(+)} \right)^{-1}. \quad (171)$$

The remaining even harmonic-oscillator states are unaffected by V_N . Consequently,

$$\sigma_+(H_N) = \sigma(B_N^{(+)}) \cup \{4N+1, 4N+5, 4N+9, \dots\}. \quad (172)$$

After ordering the elements of this union,

$$E_{n,+}^{(N)} \leq E_{n,+}, \quad (173)$$

so every eigenvalue of the intermediate Hamiltonian gives a lower bound for the corresponding exact even-parity eigenvalue of $P^2 + X^2 + X^4 + 1$.

The entire lower-bound calculation therefore reduces to three finite steps:

$$G_N^{(+)} \longrightarrow \left(G_N^{(+)} \right)^{-1} \longrightarrow \text{Eig} \left[D_N^{(+)} + \left(G_N^{(+)} \right)^{-1} \right]. \quad (174)$$

No truncation of the original Hamiltonian is involved in the inequality: the finite matrix arises from replacing the positive operator V by a finite-rank operator V_N satisfying $V_N \leq V$. The resulting eigenvalues are therefore lower bounds rather than ordinary finite-basis Rayleigh–Ritz upper bounds.

4.5 Local energy methods

Local energy methods are most naturally defined in the coordinate representation.

Lets take a standard time-independent Schrödinger equation:

$$-\frac{1}{2}\Delta\psi(x) + V(x)\psi(x) = E\psi(x) \quad (175)$$

For any trial wave function $\psi^\bullet$ and a trial energy $E^\bullet$ we can define a potential for which the $\psi^\bullet, E^\bullet$ guess is an exact solution:

$$V^\bullet(x) = \frac{1}{2} \frac{\Delta\psi^\bullet}{\psi^\bullet} + E^\bullet \quad (176)$$

And a compatible Hamilton operator looks the following:

$$H^\bullet = -\frac{1}{2}\Delta + V^\bullet \quad (177)$$

The certification idea is simple: if two self adjoint (or normal) operators differ only by a bounded operator:

$$A_1 - A_2 = B, \quad \|B\| \in \mathbb{R} \quad (178)$$

then the spectrum of A_1 has to lie inside A_2 's spectrum extended by $\|B\|$ (and vica versa). Formally:

$$\forall \lambda_2 \in \sigma(A_2), \exists \lambda_1 \in \sigma(A_1), \text{ such that } |\lambda_2 - \lambda_1| \leq \|B\| \quad (179)$$

Proof

We will use the identity:

$$A_1 - \mu I = (A_2 - \mu I)(I + (A_2 - \mu I)^{-1}(A_1 - A_2)) \quad (180)$$

$$(A_2 - \mu I)(I + (A_2 - \mu I)^{-1}(A_1 - A_2)) = (A_2 - \mu I) + (A_1 - A_2) = A_1 - \mu I \quad (181)$$

For self adjoint operators

$$\|(A - \mu I)^{-1}\| = \sup_{\lambda \in \sigma(A)} |(\mu - \lambda)^{-1}| = \left(\inf_{\lambda \in \sigma(A)} |\mu - \lambda| \right)^{-1} \quad (182)$$

So in particular if $\mu = \lambda_1 \in \sigma(A_1)$, then

$$\|(A_2 - \lambda_1 I)^{-1}\| = \left(\inf_{\lambda_2 \in \sigma(A_2)} |\lambda_1 - \lambda_2| \right)^{-1} \quad (183)$$

Now lets assume that $\inf_{\lambda_2 \in \sigma(A_2)} |\lambda_1 - \lambda_2|$ is greater then $\|B\|$:

Then

$$\|(A_2 - \lambda_1 I)^{-1}\| < \|B\|^{-1} \quad (184)$$

From $\|XY\| \leq \|X\| \cdot \|Y\|$ we have:

$$\|(A_2 - \lambda_1 I)^{-1}B\| < \|B\|^{-1}\|B\| = 1 \quad (185)$$

So

$$C = (A_2 - \lambda_1 I)^{-1}B, \quad \|C\| < 1 \quad (186)$$

Therefore for

$$A_1 - \lambda_1 I = (A_2 - \lambda_1 I)(I + (A_2 - \lambda_1 I)^{-1}B) = (A_2 - \lambda_1 I)(I + C) \quad (187)$$

and

$$(A_1 - \lambda_1 I)^{-1} = (I + C)^{-1}(A_2 - \lambda_1 I)^{-1} \quad (188)$$

If we assume that $\lambda_1 \notin \sigma(A_2)$ then $A_1 - \lambda_1 I$ is invertible whenever $I + C$ is invertible. If $\|C\| < 1$ then $I + C$ is invertible (for instance with a convergent Neumann series).

This would imply that $A_1 - \lambda_1 I$ is invertible, but this is a contradiction because $\lambda_1 \in \sigma(A_1)$.

This completes the proof.

So if we can find a $\psi^\bullet$ that induces a compatible potential $V^\bullet$ that is close to V (in supremum norm), then this guarantees that there is a spectrum point close to $E^\bullet$. Formally:

$$\exists E \in \sigma(H), |E - E^\bullet| \leq \|V - V^\bullet\|_\infty \quad (189)$$

Local energy description

$$h(x) = -\frac{1}{2} \frac{\Delta \psi^\bullet(x)}{\psi^\bullet(x)} + V(x) \quad (190)$$

Essentially the same argument gives that there has to be a spectrum point between the infimum and the supremum of the local energy $h(x)$

$$\exists E \in \sigma(H), \inf_x h(x) \leq E \leq \sup_x h(x) \quad (191)$$

4.6 Hoffman-Wielandt inequality

See [13].

4.7 Action-based semiclassical matching

In the literature, such thoughts first appeared probably in 1953 [52]

Take a 1D Schrödinger equation:

$$-\frac{1}{2} \psi''(x) + V(x) \psi(x) = E \psi(x) \quad (192)$$

This can be modified to:

$$-\frac{1}{2} \frac{\psi''(x)}{\psi(x)} + V(x) = E \quad (193)$$

Assume that we know an exact wave function solution for a potential V_0 , namely ϕ with energy E_0 , where V is “close to” V_0 .

We will search for a wave function ansatz:

$$\psi^\bullet(x) = \frac{1}{\sqrt{y'(x)}} \phi(y(x)) \quad (194)$$

where $y(x)$ is assumed to be a one-to-one monotone coordinate transformation $y: \mathbb{R} \mapsto \mathbb{R}$.

$$\psi^{\bullet'}(x) = \left(\frac{1}{\sqrt{y'(x)}} \right)' \phi(y(x)) + \frac{1}{\sqrt{y'(x)}} \phi'(y(x)) y'(x) = \left(\frac{1}{\sqrt{y'(x)}} \right)' \phi(y(x)) + \sqrt{y'(x)} \phi'(y(x)) \quad (195)$$

$$\psi^{\bullet''}(x) = \left(\frac{1}{\sqrt{y'(x)}} \right)'' \phi(y(x)) + \left(\frac{1}{\sqrt{y'(x)}} \right)' \phi'(y(x)) y'(x) + \left(\sqrt{y'(x)} \right)' \phi'(y(x)) + \sqrt{y'(x)} \phi''(y(x)) y'(x) \quad (196)$$

Direct calculation shows that:

$$\left(\frac{1}{\sqrt{y'(x)}} \right)' y'(x) + \left(\sqrt{y'(x)} \right)' = 0 \quad (197)$$

Therefore

$$\psi^{\bullet''}(x) = \left(\frac{1}{\sqrt{y'(x)}} \right)'' \phi(y(x)) + \sqrt{y'(x)} y'(x) \phi''(y(x)) \quad (198)$$

And

$$\frac{\psi^{\bullet''}(x)}{\psi^\bullet(x)} = \frac{\left(\frac{1}{\sqrt{y'(x)}} \right)''}{\frac{1}{\sqrt{y'(x)}}} + (y'(x))^2 \frac{\phi''(y(x))}{\phi(y(x))} \quad (199)$$

$$\frac{\left(\frac{1}{\sqrt{y'(x)}}\right)''}{\frac{1}{\sqrt{y'(x)}}} = -2(\mathcal{S}y)(x) \quad (200)$$

Where $\mathcal{S}f$ is the Schwarzian derivative:

$$(\mathcal{S}f)(x) = \frac{f'''(x)}{f'(x)} - \frac{3}{2} \left(\frac{f''(x)}{f'(x)} \right)^2 \quad (201)$$

Therefore

$$-\frac{1}{2} \frac{\psi^{\bullet\bullet\bullet}(x)}{\psi^\bullet(x)} + V(x) - E = -(y'(x))^2 (V_0(y(x)) - E_0) + V(x) - E + (\mathcal{S}y)(x) \quad (202)$$

Now we can try to find an appropriate “gauge transformation” $y(x)$ that would make the term

$$-(y'(x))^2 (V_0(y(x)) - E_0) + V(x) - E = 0 \quad (203)$$

zero.

Rearranging the terms:

$$y'(x) = \frac{\sqrt{V(x) - E}}{\sqrt{V_0(y(x)) - E_0}} = \frac{\sqrt{E - V(x)}}{\sqrt{E_0 - V_0(y(x))}} \quad (204)$$

$$\sqrt{V_0(y) - E_0} dy = \sqrt{V(x) - E} dx \quad (205)$$

With setting $y_0 = y(x_0)$

$$\int_{y_0}^y \sqrt{V_0(\eta) - E_0} d\eta = \int_{x_0}^x \sqrt{V(\xi) - E} d\xi \quad (206)$$

Now let's assume that we can calculate the following action integrals:

$$S(x, E) = \int_{x_0}^x \sqrt{E - V(\xi)} d\xi \quad (207)$$

$$S_0(y, E_0) = \int_{y_0}^y \sqrt{V_0(\eta) - E_0} d\eta \quad (208)$$

Then

$$S_0(y, E_0) = S(x, E) \quad (209)$$

$$y(x) = S_0^{(-1)}(S(x, E), E_0) \quad (210)$$

Because by using the derivative of an inverse and the chain rule, we get:

$$y'(x) = \frac{1}{S_0'(y(x), E_0)} S'(x, E) \quad (211)$$

It is worth mentioning three cases where this approach is applicable.

4.7.1 WKB approximation

We formally get back the Wentzel–Kramers–Brillouin approximation if we use the constant $V_0 = 0$ trivial potential as a reference system:

$$S_0(x, E_0) = x\sqrt{E_0} \quad (212)$$

$$\phi(x) = Ae^{i\sqrt{E_0}x} \quad (213)$$

$$y(x) = \frac{1}{\sqrt{E_0}} \int_0^x \sqrt{E - V(x)} dx \quad (214)$$

$$\psi^\bullet(x) = \frac{E_0^{1/4}}{(E - V(x))^{1/4}} Ae^{iS(x, E)/\sqrt{E_0}} \quad (215)$$

For instance, if we have a periodic potential on a compactified circle S^1 , then this formula could be used to approximate left- and right-moving eigenstates.

4.7.2 Airy approximation

$$V_0 = cx$$

4.7.3 Hermite approximation

Predominantly for potentials with bound states.

4.8 Concavity properties

See [24].

Lemma

For any function $f : X \times X \mapsto \mathbb{R}$

$$\inf_{x,y \in X} f(x,y) \leq \inf_{x \in X} f(x,x) \quad (216)$$

Let A and B two self adjoint matrices and $t \in [0, 1]$ ($\bar{t} = 1 - t$)

$$\lambda_1(tA + \bar{t}B) \geq t \lambda_1(A) + \bar{t} \lambda_1(B) \quad (217)$$

Proof

$$\lambda_1(tA + \bar{t}B) = \min_{v \in V, \|v\|=1} v^* \cdot (tA + \bar{t}B) \cdot v = \min_{v \in V, \|v\|=1} (t v^* \cdot A \cdot v + \bar{t} v^* \cdot B \cdot v) \quad (218)$$

$$\min_{v \in V, \|v\|=1} (t v^* \cdot A \cdot v + \bar{t} v^* \cdot B \cdot v) \geq \min_{v \in V, \|v\|=1} \min_{w \in V, \|w\|=1} (t v^* \cdot A \cdot v + \bar{t} w^* \cdot B \cdot w) = \min_{v \in V, \|v\|=1} t v^* \cdot A \cdot v + \min_{w \in V, \|w\|=1} \bar{t} w^* \cdot B \cdot w \quad (219)$$

$$\min_{v \in V, \|v\|=1} t v^* \cdot A \cdot v + \min_{w \in V, \|w\|=1} \bar{t} w^* \cdot B \cdot w = t \lambda_1(A) + \bar{t} \lambda_1(B) \quad (220)$$

The sum of the smallest eigenvalues is concave (because of the trace).

4.9 Semidefinite constrain method

This method is capable of giving a lower bound for the ground-state energy [38, 5]⁸:

$$E_0 = \inf_{\rho \geq 0, \text{Tr}(\rho)=1} \text{Tr}(\rho H) \quad (221)$$

The positivity – or more precisely the positive semidefiniteness – of the density matrix ρ means that the expectation of any positive operator (with respect to ρ) is positive:

Lemma:

$$\rho \geq 0 \iff \forall \mathcal{O} \in L(V, V), \text{Tr}(\rho \mathcal{O}^* \mathcal{O}) \geq 0 \quad (222)$$

Proof: If

$$\forall \mathcal{O} \in L(V, V), \text{Tr}(\rho \mathcal{O}^* \mathcal{O}) \geq 0 \implies \forall v \in V, v^* \cdot \rho \cdot v \geq 0 \quad (223)$$

Simply $\mathcal{O}_v = v \circ v$ (or $\mathcal{O}_v = |v\rangle\langle v|$) the set of $\mathcal{O}_v$ is a subset of all linear operators, and

$$\text{Tr}(\rho \mathcal{O}_v^* \mathcal{O}_v) = \text{Tr}(\mathcal{O}_v \rho \mathcal{O}_v^*) = \text{Tr}(|v\rangle\langle v| \rho |v\rangle\langle v|) = \langle v | \rho | v \rangle \quad (224)$$

For an orthonormal basis $e_1, \dots, e_n$:

$$\text{Tr}(A) = \sum_i \langle e_i | A | e_i \rangle \quad (225)$$

$$\forall v \in V, v^* \cdot \rho \cdot v \geq 0 \implies \forall \mathcal{O} \in L(V, V), \text{Tr}(\rho \mathcal{O}^* \mathcal{O}) \geq 0 \quad (226)$$

$$\text{Tr}(\rho \mathcal{O}^* \mathcal{O}) = \text{Tr}(\mathcal{O} \rho \mathcal{O}^*) = \sum_i \langle e_i | \mathcal{O} \rho \mathcal{O}^* | e_i \rangle \quad (227)$$

⁸For the record, the concept was first found by one of the authors on Physics Stack Exchange.

$$|f_i\rangle = \mathcal{O}|e_i\rangle \quad (228)$$

$$\text{Tr}(\rho \mathcal{O}^* \mathcal{O}) = \sum_i \langle f_i | \rho | f_i \rangle \geq 0 \quad (229)$$

The semidefinite constraint method is based on the idea that, instead of requiring positivity for all operators on a Hilbert space, we require positivity only for a selected subset of operators.

In practice, we can define a restricted vector space of operators $\mathcal{V} = \text{Span}\{\mathcal{O}_1, \dots, \mathcal{O}_K\}$ and relax the full positivity constraint to only positive operators generated from $\mathcal{V}$.

We will call a ρ' operator positive semidefinite on $\mathcal{V}$, if

$$\rho' \succ_{\mathcal{V}} 0 \iff \forall \mathcal{O} \in \mathcal{V}, \text{Tr}(\rho' \mathcal{O}^* \mathcal{O}) \geq 0 \quad (230)$$

This positivity restriction for linear operators ρ' is less restrictive than the full positivity criteria; therefore, an infimum on this larger set of linear density-like operators provides a lower bound for the smallest eigenvalue:

$$\check{E}_0 = \inf_{\rho' \succ_{\mathcal{V}} 0, \text{Tr}(\rho')=1} \text{Tr}(\rho' H) \leq \inf_{\rho \geq 0, \text{Tr}(\rho)=1} \text{Tr}(\rho H) = E_0 \quad (231)$$

Lemma: $\mathcal{O}_{\underline{a}} = \sum_k a_k \mathcal{O}_k \in \mathcal{V}$

$$\forall a \in \mathbb{R}^K, \text{Tr}(\rho' \mathcal{O}_{\underline{a}}^* \mathcal{O}_{\underline{a}}) \geq 0 \iff A \succ 0, A_{ij} = \text{Tr}(\rho' \mathcal{O}_i^* \mathcal{O}_j) \quad (232)$$

Proof:

$$\text{Tr}(\rho' \mathcal{O}_{\underline{a}}^* \mathcal{O}_{\underline{a}}) = \sum_{i,j} \bar{a}_i a_j \text{Tr}(\rho' \mathcal{O}_i^* \mathcal{O}_j) = \sum_{i,j} \bar{a}_i a_j A_{ij} = \underline{a}^* \cdot A \cdot \underline{a} \quad (233)$$

Therefore, if the finite A matrix, created from the expectations of the combinations of basis operators, is positive semidefinite, then $\rho' \succ_{\mathcal{V}} 0$.

If the Hamiltonian operator H is constructible as a linear combination of basis operator pairs, i.e. $H = \sum_{i,j} \alpha_{i,j} \mathcal{O}_i^* \mathcal{O}_j$, then the minimization with the relaxed semidefiniteness criteria can be reformulated to the language of operator expectations.

If the set of basis operators contain the identity, then the whole constrained minimization process can be reformulated on the language of operator expectations $\langle \mathcal{O}^* \mathcal{O} \rangle \in \mathbb{C}$ – i.e. a finite set of real complex numbers – without referring to Hilbert spaces or density operators. (This is inline with the spirit of Heidenberg's approach from 1925, [32, 77, 33]⁹)

An elementary example is the quantum harmonic oscillator:

$$H = \frac{1}{2} P^2 + \frac{1}{2} X^2 \quad (234)$$

$$\mathcal{O} = I, \quad \mathcal{O}_1 = X, \quad \mathcal{O}_2 = P \quad (235)$$

$$A = \begin{bmatrix} \langle I \rangle & \langle X \rangle & \langle P \rangle \\ \langle X \rangle & \langle X^2 \rangle & \langle XP \rangle \\ \langle P \rangle & \langle PX \rangle & \langle P^2 \rangle \end{bmatrix} \quad (236)$$

$$\langle H \rangle = \frac{1}{2} \langle P^2 \rangle + \frac{1}{2} \langle X^2 \rangle \quad (237)$$

And from the canonical commutation relations:

$$[X, P] = iI \implies \langle XP \rangle = \frac{1}{2} \langle XP + PX \rangle + \frac{1}{2} i \langle I \rangle \quad (238)$$

and $\langle PX \rangle = \frac{1}{2} \langle XP + PX \rangle - \frac{1}{2} i \langle I \rangle$

$$A \succ 0 \implies \langle X^2 \rangle \langle P^2 \rangle - \langle XP \rangle \langle PX \rangle = \langle X^2 \rangle \langle P^2 \rangle - \frac{1}{4} (\langle \{X, P\} \rangle^2 + 1) \geq 0 \quad (239)$$

$$\langle X^2 \rangle \langle P^2 \rangle \geq \frac{1}{4} (\langle \{X, P\} \rangle^2 + 1) \geq \frac{1}{4} \quad (240)$$

That is explicitly the Heisenberg uncertainty principle.

⁹For more about the Umdeutung or reinterpretation paper see this neatly animated presentation by Dr. Jorge S. Diaz.

$$\langle X^2 \rangle \geq \frac{1}{4} \frac{1}{\langle P^2 \rangle} \implies \langle H \rangle \geq \frac{1}{2} \langle P^2 \rangle + \frac{1}{8} \frac{1}{\langle P^2 \rangle} \quad (241)$$

$$\tilde{E}_0 = \inf_{\langle P^2 \rangle > 0} \frac{1}{2} \langle P^2 \rangle + \frac{1}{8} \frac{1}{\langle P^2 \rangle} = \frac{1}{2} \quad (242)$$

Where the infimum is reached at $\langle P^2 \rangle = 1/2$.

And the corresponding positive definite A_* matrix looks:

$$A_* = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2}i \\ 0 & -\frac{1}{2}i & \frac{1}{2} \end{bmatrix} \quad (243)$$

Quartic oscillator

$$H = \frac{1}{2} P^2 + X^4 \quad (244)$$

$$\mathcal{O} = I, \quad \mathcal{O}_1 = P, \quad \mathcal{O}_2 = X, \quad \mathcal{O}_3 = X^2 \quad (245)$$

$$A = \begin{bmatrix} \langle I \rangle & \langle P \rangle & \langle X \rangle & \langle X^2 \rangle \\ \langle P \rangle & \langle P^2 \rangle & \langle PX \rangle & \langle PX^2 \rangle \\ \langle X \rangle & \langle XP \rangle & \langle X^2 \rangle & \langle X^3 \rangle \\ \langle X^2 \rangle & \langle X^2 P \rangle & \langle X^3 \rangle & \langle X^4 \rangle \end{bmatrix} \quad (246)$$

$$\langle H \rangle = \frac{1}{2} \langle P^2 \rangle + \frac{1}{2} \langle X^4 \rangle \quad (247)$$

$$A \geq 0 \implies \langle X^2 \rangle \langle P^2 \rangle - \langle XP \rangle \langle PX \rangle = \langle X^2 \rangle \langle P^2 \rangle - \frac{1}{4} (\langle \{X, P\} \rangle^2 + 1) \geq 0 \quad (248)$$

$$A \geq 0 \implies \langle I \rangle \langle X^4 \rangle - \langle X^2 \rangle \langle X^2 \rangle \geq 0 \implies \langle X^4 \rangle \geq \langle X^2 \rangle^2 \quad (249)$$

$$\langle H \rangle \geq \frac{1}{2} \langle P^2 \rangle + \left(\frac{1}{4} \frac{1}{\langle P^2 \rangle} \right)^2 \quad (250)$$

$$\tilde{E}_0 = \inf_{\langle P^2 \rangle > 0} \frac{1}{2} \langle P^2 \rangle + \left(\frac{1}{4} \frac{1}{\langle P^2 \rangle} \right)^2 = \frac{3}{4^{4/3}} \approx 0.472 \quad (251)$$

Where the infimum is reached at $\langle P^2 \rangle = 2^{-2/3}$.

$$A_* = \begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & i\frac{1}{2} & 0 \\ 0 & -i\frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{4} \end{bmatrix} \quad (252)$$

That is a positive semidefinite matrix with eigenvalues $(0, 0, 1, 5/4)$

Generalisation to compute more accurate lower bounds

Weyl - ordered monomial product [25]:

$$\mathcal{W}_{n,m} = \frac{1}{(n+m)!} \sum_{\pi \in S(n+m)} A_{\pi(1)} \cdots A_{\pi(n+m)}, \quad A_1 = \cdots = A_n = x, \quad A_{n+1} = \cdots = A_{n+m} = p.$$

With a product expansion:

$$\mathcal{W}_{n,m} \mathcal{W}_{r,s} = \sum_{k=0}^{\min(m,r)} \sum_{\ell=0}^{\min(n,s)} \left(\frac{i}{2} \right)^{k+\ell} (-1)^\ell k! \ell! \binom{m}{k} \binom{r}{k} \binom{n}{\ell} \binom{s}{\ell} \mathcal{W}_{n+r-k-\ell, m+s-k-\ell}.$$

From these operators, one can systematically build finite A matrices containing basis operators $\{\mathcal{W}_{n,m} | n, m \geq 0, n+m \leq M\}$. If the Hamiltonian H can be constructed from a linear combination of $\mathcal{W}$ pair products, then a numerical semidefinite optimisation can provide a rigorous lower bound; `SemidefiniteOptimization` can be used in Wolfram Mathematica.

4.10 Wigner function and phase space formalism

For the harmonic oscillator

$$W_n(I) = \frac{(-1)^n}{\pi} L_n(4I) e^{-2I} \quad (253)$$

For a Hamiltonian operator H , construct its phase space function $H(x, p)$, determine the Action-angle variables (I, φ) , and then we can make an ansatz for the density matrix:

$$R_n = W_n(I) \star W_n(I) \quad (254)$$

on the phase space. Where $\star$ represents the Moyal product.

We need to square the original $W_n(\cdot)$ functions, because otherwise the positivity of the phase-space analogue of the density matrix is not guaranteed.

Of course, this is only an ansatz; other validation methods are needed to give exact error bounds for the eigenenergies related to this guess. One seemingly feasible option is to compute the Krylov-Weinstein bounds.

5 Matrix elements and Form factors

For a self-adjoint operator A , we can define a perturbed Hamiltonian operator:

$$H(\kappa) = H + \kappa A \quad (255)$$

If, for some interval $\kappa \in (\underline{\kappa}, \bar{\kappa})$ containing 0, $H(\kappa)$ is bounded from below, then we can derive an exact bound for the ground state matrix element of A with respect to the ground state.

From perturbation theory, for eigenvalues and eigenvectors:

$$H|\psi_n\rangle = E_n|\psi_n\rangle \quad (256)$$

$$H(\kappa)|\psi_n(\kappa)\rangle = E_n(\kappa)|\psi_n(\kappa)\rangle \quad (257)$$

$$E'_n(0) = \langle \psi_n(0) | A | \psi_n(0) \rangle \quad (258)$$

5.1 Form factors for vacuum or ground state

On the other hand

$$E_0(\kappa) = \inf_{\psi \in \mathcal{H}, \|\psi\|=1} \langle \psi | H(\kappa) | \psi \rangle \quad (259)$$

from which follows that $E_0(\kappa)$ is a concave function. (In fact $\sum_{k=0}^n E_k(\kappa)$ is also concave.)

From three upper and lower bounds for the perturbed ground-state energy for $\underline{\kappa} \leq \kappa_- < 0 < \kappa_+ \leq \bar{\kappa}$ We can give exact lower and upper bounds for the ground state matrix element (or form factor of A):

$$\frac{E_0(\kappa_+) - \overline{E}_0(0)}{\kappa_+} \leq E'_0(0) \leq \frac{E_0(\kappa_-) - \overline{E}_0(0)}{\kappa_-} \quad (260)$$

In fact, for the exact bound we need only an upper bound for E_0 , $\overline{E}_0$ and two lower bounds for the ground states at κ_- and κ_+ , $\underline{E}_0(\kappa_-)$ and $\underline{E}_0(\kappa_+)$

For a non-self-adjoint operator $\mathcal{O}$ it can be split to two self-adjoint operators $A = (\mathcal{O} + \mathcal{O}^*)/2$, $B = (\mathcal{O} - \mathcal{O}^*)/(2i)$ with $\mathcal{O} = A + iB$

5.2 “Large operators”

If $H + \kappa A$ would become unbounded from below if $\kappa < 0$ – consider, for example, $A = X^4$ for a harmonic oscillator – but for some power of H the expression would stay bounded, then we can define the following operator:

$$F(\kappa) = (H - E)^m + \kappa A \quad (261)$$

If $E < E_0$, the same arguments apply to the lowest eigenvalue of $F(\kappa)$, which provides exact bounds for the vacuum form factors of A .

5.3 Diagonal form factors for excited states

$$F(\kappa) = (H - E)^2 + \kappa A \quad (262)$$

If E is close to E_n , the n -th excited energy level, then by the same convexity argument, we can derive exact bounds for the diagonal form factor of A with respect to the n -th excited state ψ_n .

5.4 Off-diagonal Form factors

We want to solve the following problem. We want to derive exact bounds for the absolute value of the matrix element $|a_{nm}| = |\langle \psi_n | A | \psi_m \rangle|$ for a self-adjoint operator A where ψ_n and ψ_m are exact eigenstates of the Hamiltonian H with energies E_n and E_m .

(Eigenstates have phase freedom in complex Hilbert spaces; therefore, the full a_{nm} is not “gauge-independent”. Taking the absolute value of the matrix element provides a well-defined, gauge-independent quantity.)

First we define a doubled Hilbert space:

$$\mathcal{H}^2 = \mathcal{H} \otimes \mathcal{H} \quad (263)$$

And a separable squared Hamilton operator:

$$F(E_n^\bullet, E_m^\bullet) = (H - E_n^\bullet)^2 \otimes I + I \otimes (H - E_m^\bullet)^2 \quad (264)$$

Where $E_n^\bullet$ and $E_m^\bullet$ are approximate guesses for the exact E_n, E_m eigenvalues.

Assuming that the n -th and m -th eigenstates are non-degenerate, the ground state of this operator is exactly $|\Psi_0\rangle = |\psi_n\rangle \otimes |\psi_m\rangle$ (with ground-state eigenvalue $(E_n - E_n^\bullet)^2 + (E_m - E_m^\bullet)^2$)

Now we can introduce a perturbation operator that mixes states from the two separate Hilbert spaces:

$$\mathcal{A} = (A \otimes A) \mathcal{C} \quad (265)$$

Where $\mathcal{C}$ is a cyclic operator:

$$\mathcal{C}|\varphi_1\rangle \otimes |\varphi_2\rangle = |\varphi_2\rangle \otimes |\varphi_1\rangle \quad (266)$$

For example for $\mathcal{H} = L^2(\mathbb{R}, \mathbb{C})$, $\mathcal{H}^2 = L^2(\mathbb{R}^2, \mathbb{C})$, and $\mathcal{C}\psi(x, y) = \psi(y, x)$

The cyclic operator commutes with the symmetric $A \otimes A$ operator:

$$(A \otimes A) \mathcal{C}|\varphi_1\rangle \otimes |\varphi_2\rangle = (A|\varphi_2\rangle) \otimes (A|\varphi_1\rangle) = \mathcal{C} (A \otimes A)|\varphi_1\rangle \otimes |\varphi_2\rangle \quad (267)$$

Therefore $\mathcal{A}$ is self-adjoint. $(\mathcal{A}^\dagger = \mathcal{C} (A^\dagger \otimes A^\dagger) = (A \otimes A) \mathcal{C} = \mathcal{A})$

The ground-state expectation for $\mathcal{A}$ in $\mathcal{H}^2$ is:

$$\langle \Psi_0 | \mathcal{A} | \Psi_0 \rangle = \langle \psi_n | A | \psi_m \rangle \langle \psi_m | A | \psi_n \rangle = |a_{nm}|^2 \quad (268)$$

By introducing:

$$F(\kappa; E_n^\bullet, E_m^\bullet) = (H - E_n^\bullet)^2 \otimes I + I \otimes (H - E_m^\bullet)^2 + \kappa \mathcal{A} \quad (269)$$

From perturbation theory, we get:

$$F'_0(0) = \frac{\partial}{\partial \kappa} \lambda_0(F(\kappa; E_n^\bullet, E_m^\bullet))|_{\kappa=0} = \langle \Psi_0 | \mathcal{A} | \Psi_0 \rangle = |a_{nm}|^2 \quad (270)$$

Again, using the concavity of the ground-state energy of self-adjoint operators, we can obtain an exact estimate for $|a_{nm}|$.

If we have exact lower estimates for finite $\kappa_- < 0 < \kappa_+$ values and an exact upper estimate for the lowest eigenvalue for $\kappa = 0$, then we can provide the same upper and lower bounds for the first-order lowest eigenstate derivative:

$$\frac{F_0(\kappa_+) - \overline{F_0}(0)}{\kappa_+} \leq F'_0(0) \leq \frac{F_0(\kappa_-) - \overline{F_0}(0)}{\kappa_-} \quad (271)$$

$$\sqrt{\max\left(0, \frac{F_0(\kappa_+) - \overline{F_0}(0)}{\kappa_+}\right)} \leq |a_{nm}| \leq \sqrt{\frac{F_0(\kappa_-) - \overline{F_0}(0)}{\kappa_-}} \quad (272)$$

5.5 Full complex Form factors for dense operators

The phase of a non-diagonal matrix element between different eigenstates is not well defined on a complex Hilbert space, because the eigenstates have phase freedom. $\psi'_n = e^{i\alpha_n}\psi_n$ is also an eigenstate with the same eigenenergy.

Therefore, the matrix element $\langle\psi_n|A|\psi_m\rangle$ changes if we change the phases of the eigenstates. $\langle\psi'_n|A|\psi'_m\rangle = e^{i(\alpha_m - \alpha_n)}\langle\psi_n|A|\psi_m\rangle$.

However, for any fixed choice of the sets of eigenstates, the complex matrix elements will correspond to a representative from an equivalence class $D^\dagger AD$, for some diagonal phase transformation $D_{nm} = \delta_{nm}e^{i\alpha_n}$. (We will call D a specific gauge.)

If we fix a “reality condition” on A , i.e. we choose a gauge where, for instance, the r -th row contains real matrix elements, and the A matrix is dense enough (here we assume that all matrix elements are non-zero), then one can recover the other complex matrix elements in the following way:

First we can introduce “gauge-independent cyclic triple products”

$$a(i, j, k) = \langle i|A|j\rangle\langle j|A|k\rangle\langle k|A|i\rangle \quad (273)$$

This can be a complex number in general, but its real and imaginary parts can be defined as form factors on a tripled Hilbert space $\mathcal{H}^{\otimes 3} = \mathcal{H} \otimes \mathcal{H} \otimes \mathcal{H}$

$$F(E_i^\bullet, E_j^\bullet, E_k^\bullet) = (H - E_i^\bullet)^2 \otimes I \otimes I + I \otimes (H - E_j^\bullet)^2 \otimes I + I \otimes I \otimes (H - E_k^\bullet)^2 \quad (274)$$

and two self-adjoint perturbation operators

$$\mathcal{A}_{\Re} = \frac{1}{2} \left((A \otimes A \otimes A) \mathcal{C} + \mathcal{C}^\dagger (A \otimes A \otimes A) \right) \quad (275)$$

$$\mathcal{A}_{\Im} = \frac{1}{2i} \left((A \otimes A \otimes A) \mathcal{C} - \mathcal{C}^\dagger (A \otimes A \otimes A) \right) \quad (276)$$

Where $\mathcal{C}$ is a cyclic operator shifting the variables one to the left:

$$\mathcal{C}|i\rangle \otimes |j\rangle \otimes |k\rangle = |j\rangle \otimes |k\rangle \otimes |i\rangle \quad (277)$$

$$\mathcal{C}^\dagger|i\rangle \otimes |j\rangle \otimes |k\rangle = |k\rangle \otimes |i\rangle \otimes |j\rangle \quad (278)$$

The ground state of $F(E_i^\bullet, E_j^\bullet, E_k^\bullet)$ is $|\Psi_0\rangle = |i\rangle \otimes |j\rangle \otimes |k\rangle$

And

$$\langle\Psi_0|A_{\Re}|\Psi_0\rangle = \frac{1}{2} (a_{ij}a_{jk}a_{ki} + (a_{ij}a_{jk}a_{ki})^*) = \Re(a_{ij}a_{jk}a_{ki}) \quad (279)$$

$$\langle\Psi_0|A_{\Im}|\Psi_0\rangle = \frac{1}{2i} (a_{ij}a_{jk}a_{ki} - (a_{ij}a_{jk}a_{ki})^*) = \Im(a_{ij}a_{jk}a_{ki}) \quad (280)$$

Therefore, we can introduce:

$$F_{\Re}(\kappa; E_i^\bullet, E_j^\bullet, E_k^\bullet) = F(E_i^\bullet, E_j^\bullet, E_k^\bullet) + \kappa \mathcal{A}_{\Re} \quad (281)$$

$$F_{\Im}(\kappa; E_i^\bullet, E_j^\bullet, E_k^\bullet) = F(E_i^\bullet, E_j^\bullet, E_k^\bullet) + \kappa \mathcal{A}_{\Im} \quad (282)$$

And from this, we can get exact upper and lower bounds for the real and imaginary parts of $a(i, j, k) = a_{ij}a_{jk}a_{ki}$

Now, if we are interested in the complex value of a_{jk} in a gauge where all a_{sj} matrix elements are defined to be real and positive (this can be done if the operator is dense in the energy eigenstate basis), then we have the simple formula:

$$a_{jk} = \frac{a(s, j, k)}{\sqrt{|a_{sj}|^2} \sqrt{|a_{ks}|^2}} \quad (283)$$

For the absolute value squares of matrix elements, we derived exact upper and lower bounds in the previous section; therefore, if we know lower bounds greater than 0 for $|a_{sj}|^2$ and $|a_{ks}|^2$, then we have exact upper and lower bounds for the real and imaginary parts of a_{jk} .

The cyclic products introduced above are operator-inserted analogues of the Bargmann invariants. Like the usual n -vertex Bargmann invariants, they are invariant under independent rephasings of the participating states [54, 4].

Remark. The assumption that all matrix elements of A are non-zero is stronger than necessary. Consider the support graph of A , whose vertices are the energy eigenstates and whose edges correspond to non-zero matrix elements a_{ij} , in analogy with the directed-graph formulation discussed for Brualdi's theorem in Section 2.2 [15]. For the construction above, it is enough that the relevant matrix elements form a triangle, so that the gauge-invariant product $a_{ij}a_{jk}a_{ki}$ is non-zero. More generally, any closed cycle $i_1 \rightarrow \dots \rightarrow i_n \rightarrow i_1$ gives the gauge-invariant product $a_{i_1 i_2} \dots a_{i_n i_1}$. Thus, A need not be dense; it is sufficient for its support graph to be sufficiently rich in cycles for the matrix elements of interest.

6 Quantum Mechanical Bootstrap

Quantum mechanical bootstrap, first introduced in [31], is a framework for obtaining exact solutions to quantum mechanical systems by imposing consistency conditions that constrain the space of allowed solutions. While the original work focused on the quantum mechanics of large- N matrix models, subsequent developments have demonstrated that these techniques can be extended to derive non-perturbative results for more familiar one-dimensional systems, such as the anharmonic oscillator [12].

In this work, we first construct the bootstrap method and then apply it to the simple harmonic oscillator as a benchmark to demonstrate its accuracy, before extending the analysis to more complex systems that are typically accessible only through perturbative approaches.

6.1 The Operator Spectrum as a Moment Problem

In Quantum Mechanics, we are usually interested in calculating the expectation values of operators. Expectation values are usually the only "measurable" quantities of a theory.

We define the expectation value of an operator $\mathcal{O}$ as

$$\langle \mathcal{O} \rangle = \langle \psi | \mathcal{O} | \psi \rangle. \quad (284)$$

Considering the resolution of the identity,

$$\mathbb{K} = \int_{-\infty}^{+\infty} dx |x\rangle \langle x|, \quad (285)$$

and inserting Eq. (285) on either side of $\mathcal{O}$ in Eq. (284), we obtain

$$\langle \mathcal{O} \rangle = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} dx dy \langle \psi | x \rangle \langle x | \mathcal{O} | y \rangle \langle y | \psi \rangle. \quad (286)$$

Taking,

$$\mathcal{O} = f(\hat{x})$$

We get,

$$\langle x | \mathcal{O} | y \rangle = f(x) \delta(x - y)$$

Thus, Eq. (284) can be formulated as

$$\langle \mathcal{O} \rangle = \int_{-\infty}^{+\infty} dx \psi^*(x) f(x) \psi(x) \quad (287)$$

Or,

$$\langle \mathcal{O} \rangle = \int_{-\infty}^{+\infty} dx f(x) |\psi(x)|^2 \quad (288)$$

Consider the following case,

$$f(x) = x^n \quad (289)$$

It's easy to see that Eq. (288) can be written as,

$$s_n = \langle x^n \rangle = \int_{-\infty}^{+\infty} x^n d\mu \quad (290)$$

Where,

$$d\mu = |\psi(x)|^2 dx \quad (291)$$

We refer to μ as the measure and to Eq. (290) as the "Hamburger Moment Problem" (citation needed).

Integrating both sides of Eq.(291), it is obvious to see that

$$\mu = \int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1 \quad (292)$$

Thus, the measure is a "positive measure" and the L.H.S of Eq. (290) is a sequence of positive moments. In other words, Eq. (290) is solvable if the Hankle matrix defined by,

$$M_{ij} = s_{i+j}$$

is Positive Semidefinite $M \succeq 0$. Where, $0 \leq i, j \leq N$

6.2 Construction of the Quantum Mechanical Bootstrap

In the language of 6.1, solving a quantum mechanical problem can be described as finding a unique positive measure such that the moment sequence converges for any set of Operators [57].

We start by closely following the construction described in [12].

Consider the Time Independent Schrodinger equation,

$$\hat{H} |\psi\rangle = E |\psi\rangle \quad (293)$$

Where,

$$\hat{H} = -\frac{1}{2m} \hat{p}^2 + V(\hat{x})$$

For any operator $\mathcal{O}$, we have the following relation in a energy Eigenstate,

$$\langle [H, \mathcal{O}] \rangle = 0 \quad (294)$$

Thus we have,

$$\langle H\mathcal{O} \rangle = E\langle \mathcal{O} \rangle \quad (295)$$

From, equations (294) and (295), it is clear that we can construct a closed set of equations that tightly constrain the allowed values that the energy can take. We can arrive at a simple but rich set of equations by choosing an appropriate set of operators.

For polynomial potentials this choice was presented in [12].

We consider the following set of operators,

$$\begin{aligned} \mathcal{O}_1 &= x^n p \\ \mathcal{O}_2 &= x^{n-1} \end{aligned}$$

From,

$$\langle [H, \mathcal{O}_1] \rangle = -\frac{1}{2m} [\hat{p}^2, x^n p] + [V(\hat{x}), x^n p] \quad (296)$$

Using the relation

$$[A, BC] = [A, B]C + B[A, C] \quad (297)$$

We rewrite

$$\langle [H, \mathcal{O}_1] \rangle = -\frac{1}{2m} [\hat{p}^2, x^n] p + x^n [V(\hat{r}), p] \quad (298)$$

Taking,

$$\hat{p} = -i \frac{\partial}{\partial x} \quad (299)$$

and,

$$[\hat{x}, \hat{p}] = i \quad (300)$$

We get,

$$[V(\hat{x}), \hat{p}] = iV'(x) \quad (301)$$

and

$$[\hat{x}^n, \hat{p}] = inx^{n-1} \quad (302)$$

$$[\hat{p}^2, x^n] = \hat{p}[\hat{p}, \hat{x}^n] + [\hat{p}, \hat{x}^n]\hat{p} = (-inx^{n-1})\hat{p} + \hat{p}(-inx^{n-1}) \quad (303)$$

Which results in,

$$\hat{p}[\hat{p}, \hat{x}^n] + [\hat{p}, \hat{x}^n]\hat{p} = -2inx^{n-1}\hat{p} - n(n-1)x^{n-2} \quad (304)$$

Thus,

$$[\hat{p}^2, x^n] = -2inx^{n-1}\hat{p}^2 - n(n-1)x^{n-2}\hat{p} \quad (305)$$

Which gives us,

$$\langle [H, \mathcal{O}_1] \rangle = -\frac{in}{m} \langle x^{n-1}\hat{p}^2 \rangle - \frac{n(n-1)}{2m} \langle x^{n-2}\hat{p} \rangle + i \langle x^n V'(x) \rangle \quad (306)$$

Thus,

$$\frac{in}{m} \langle x^{n-1}\hat{p}^2 \rangle + \frac{n(n-1)}{2m} \langle x^{n-2}\hat{p} \rangle - i \langle x^n V'(x) \rangle = 0 \quad (307)$$

Eq. (307) enables us to recursively calculate higher moments which make up the Hankle matrix. Eq. (307) contains a mix of $\langle \hat{x} \rangle$ and $\langle \hat{p} \rangle$. In order to rewrite it purely in terms of $\langle \hat{x} \rangle$ and to involve energy into the mix, we do the following.

Let's take $\mathcal{O}_2$ and insert it into Eq. (295).

This gives us,

$$\frac{-1}{2m} \langle x^{n-1}\hat{p}^2 \rangle + \langle V(x)x^{n-1} \rangle = E \langle x^{n-1} \rangle \quad (308)$$

Thus, we find the following relation for the first term in Eq. (307)

$$\langle x^{n-1}\hat{p}^2 \rangle = 2m \langle V(x)x^{n-1} \rangle - 2mE \langle x^{n-1} \rangle \quad (309)$$

Let us now insert $\mathcal{O}_2$ into Eq. (294).

$$\langle [H, \mathcal{O}_2] \rangle = \langle -\frac{1}{2m} [\hat{p}^2, x^{n-1}] + [V(\hat{x}), x^{n-1}] \rangle \quad (310)$$

We immediately see that,

$$\langle [\hat{p}^2, x^{n-1}] \rangle = -2i(n-1)x^{n-2}\hat{p}^2 - (n-1)(n-2)x^{n-3}\hat{p} = 0 \quad (311)$$

Which gives us

$$\langle x^{n-2}\hat{p} \rangle = \frac{i}{2} (n-2) \langle x^{n-3} \rangle \quad (312)$$

Now we can rewrite Eq. (307) as,

$$-2in \langle V(x)x^{n-1} \rangle + 2inE \langle x^{n-1} \rangle - \frac{in(n-1)(n-2)}{4m} \langle x^{n-3} \rangle + i \langle x^n V'(x) \rangle = 0 \quad (313)$$

setting $m = 1$ we get,

$$2nE \langle x^{n-1} \rangle + \frac{n(n-1)(n-2)}{4} \langle x^{n-3} \rangle - \langle x^n V'(x) \rangle - 2n \langle x^{n-1} V(x) \rangle = 0 \quad (314)$$

6.3 Simple Harmonic Oscillator

6.4 Anharmonic Oscillator

6.5 Quantum Simple Pendulum

7 Combination of the methods

7.1 Perturbation theory upgraded by exact bounds

Perturbation theory can be supplemented by rigorous error and reminder terms when the perturbation is bounded [35].

However for many interesting problems the perturbation can be unbounded, or even “out grow” the original operator. Such case is the anharmonic oscillator with quartic term. It was proven by Bender and Wu that the perturbation series for the anharmonic oscillator diverges i.e. its radius of convergence is 0 [10, 11].

For the g -normalisation (51) with $\omega_0 = 1$, the ground-state energy has the asymptotic expansion

$$E_0(g) = \sum_{n=0}^N a_n g^n + \mathcal{O}(g^{N+1}), \quad g \rightarrow 0^+. \quad (315)$$

for every fixed N . The perturbative coefficients have the large-order behaviour

$$a_n \sim -\frac{\sqrt{6}}{\pi} \left(-\frac{3}{4}\right)^n \frac{(2n)!}{n!}, \quad n \rightarrow \infty. \quad (316)$$

Thus the formal perturbation series has zero radius of convergence.

Another worrying sign is that for $g < 0$, H_g is no longer bounded from below. This is analogous to the instability underlying Dyson’s argument for the divergence of perturbation theory in quantum electrodynamics [19]¹⁰[67, 8].

Despite these difficulties, we shall show that, by carefully combining different techniques, one can obtain rigorous and often quite accurate results for this unbounded perturbation problem.

¹⁰Although the argument is invalid as a proof of zero radius of convergence, it signals an important property of the problem. For a counterexample to the argument, consider $H(g) = P^2 + gP^4$ on the circle. For $g < 0$ the operator is unbounded from below; however, all its parameter-dependent eigenenergies have convergent perturbative series, namely $E_n(g) = n^2 + gn^4$. For a more direct application of this idea to the anharmonic oscillator, see [30].

A Extension of the function algebra with composition

For a richer function algebra we can extend the allowed symbolic combinations with function composition
 $\circ: \mathcal{A} = \langle \mathbb{R}, \cdot, \oplus, \otimes, \circ, \mathbf{a}_1, \mathbf{a}_2, \dots \rangle$

However here we need to take care of the welldefined-ness of the expressions. We introduce the boolean function $\text{Valid} : \mathcal{A} \mapsto \{\text{True}, \text{False}\}$

$$\text{Valid}[c \cdot A] = \text{Valid}[A] \quad (317)$$

$$\text{Valid}[A_1 \oplus A_2](x) = \text{Valid}[A_1 \otimes A_2](x) = \text{Valid}[A_1](x) \wedge \text{Valid}[A_2](x) \quad (318)$$

$$\overline{\text{Max}}_0^+[c \cdot A] = \begin{cases} |c| \cdot \overline{\text{Max}}_0^+[A] & \text{if } \text{Valid}[A] \\ \infty & \text{if } \neg \text{Valid}[A] \end{cases} \quad (319)$$

...Similar for plus and times...

$$\text{Valid}[A_2 \circ A_1] = \begin{cases} \text{Valid}[A_1](x) \wedge \text{Valid}[A_2](x) & \text{if } [-\overline{\text{Max}}_0^+[A_1], \overline{\text{Max}}_0^+[A_1]] \subseteq \mathcal{D} \\ \text{False} & \text{otherwise} \end{cases} \quad (320)$$

$$\overline{\text{Max}}_0^+[A_2 \circ A_1] = \begin{cases} \overline{\text{Max}}_0^+[A_2] & \text{if } \text{Valid}[A_2 \circ A_1] \\ \infty & \text{if } \neg \text{Valid}[A_2 \circ A_1] \end{cases} \quad (321)$$

If $\text{Valid}[A_2 \circ A_1]$ is True, then

$$\overline{\text{Max}}_1^+[A_2 \circ A_1] = \overline{\text{Max}}_1^+[A_2] \cdot \overline{\text{Max}}_1^+[A_1] \quad (322)$$

$$\overline{\text{Max}}_2^+[A_2 \circ A_1] = \overline{\text{Max}}_2^+[A_2] \cdot (\overline{\text{Max}}_1^+[A_1])^2 + \overline{\text{Max}}_1^+[A_2] \cdot \overline{\text{Max}}_2^+[A_1] \quad (323)$$

(If $\text{Valid}[A_2 \circ A_1]$ is False, then by convention all these bounds can be set to ∞ .)

Without introducing the concept of validity maybe only with introducing infinite upper bounds we can extend $\overline{\text{Max}}_0^+$ for the compositional algebra.

$$\overline{\text{Max}}_0^+[A_2 \circ A_1] = \begin{cases} \overline{\text{Max}}_0^+[A_2] & \text{if } [-\overline{\text{Max}}_0^+[A_1], \overline{\text{Max}}_0^+[A_1]] \subseteq \mathcal{D} \\ \infty & \text{otherwise} \end{cases} \quad (324)$$

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